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Convergence analysis of Exponential Weights algorithm in day-ahead electricity market auctions

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Student: R. Schinina

Supervisor: Prof. Dr. J. G. Peypouquet

Co-supervisor: Prof. Dr. A. Cherukuri

Abstract

In Day-Ahead energy auctions, bidders must bid strategically without prior knowledge of the market-clearing price or competitors' decision-making process. While no-regret learning algorithms provide a framework to utilize post-auction data to adjust bidding choices, their long-term bid preferences remain unidentified. In this thesis, we address this gap by studying the asymptotic behavior of auction participants using the EW algorithm based on their cost functions, capacities, and strategy sets. For homogeneous players with two bid choices, we establish explicit conditions for the lowest-bid preference and show that if the aggregate capacities of the players equal the demand, then the highest bid is preferred. Moreover, we identify submartingale to supermartingale regime shifts when the aggregate capacity exceeds demand. Lastly, we extend this analysis to heterogeneous players, proving weight convergence across multiple strategic overlap scenarios.

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1 Introduction

Day-Ahead auctions are a vital component of the energy markets. Power exchange members rely on them to trade the largest share of energy in the market (EPEX SPOT, 2025a). This thesis focuses on the bidder's perspective, assuming individual buyers' demands are aggregated into a single demand value. The primary objective of these bidders is to maximize their payoff when selling energy to the buyers. In order to achieve this, bidders must strategically price their bids low enough to ensure acceptance. While a significant portion of the literature focuses on game-theoretic approaches (implicitly assuming that bidders possess prior information regarding the market-clearing prices or rival decision-making processes (Hobbs, Metzler & Pang, 2000; Hu & Ralph, 2007; Ruiz & Conejo, 2009)), the real-world electricity markets operate under anonymity (EPEX SPOT, 2025a). In practice, no bidder has insights into the competitors' strategies beforehand, and only aggregate data is published post-auction (EPEX SPOT, 2025b). A recent study by Getaneh Abate et al. (2024) directly addresses this realistic scenario by investigating how bidding decision models can be dynamically updated using post-auction data. The authors adopt a no-regret learning perspective, focusing specifically on the Exponential Weights (EW) algorithm (a special case of the Hedge(β) algorithm proposed by Freund and Schapire (1997)), which for a learning rate equal to $\sqrt{\frac{8 \ln N}{T}}$, is known to obtain the best regret rates in online learning (Getaneh Abate et al., 2024). Their work provides a valuable preliminary evaluation of the EW algorithm's effectiveness in energy auctions.

However, as Professor Woolsey famously noted, "A manager would rather live with a problem he cannot solve than accept a solution he does not understand" (Wessels, 2015). In the context of energy auctions, it is important to be able to justify why a specific bid was selected rather than treating the algorithm as a black box. To provide the necessary interpretation, we have to understand the behavior of the algorithm across repeated auctions. Different types of convergence have been shown. In (Freund & Schapire, 1997), the authors prove that the average loss of the Hedge(β) algorithm approaches the average loss of the best strategy with a worst rate of convergence of $\mathcal{O}\left(\sqrt{\frac{\ln N}{T}}\right)$ where N is the number of strategies and T is the number of turns the game is played for. No-regret algorithms are

also related to equilibria. In particular, in (Hart & Mas-Colell, 2000) the authors show that if all players play according to a no-regret algorithm, the joint distribution of play converges to the set of coarse correlated equilibria. More recently, d'Andrea, Gensbittel and Renault (2024) extended these equilibria convergence results by proving that the limits of the EW algorithm are a special type of Nash equilibrium known as Nash Equilibrium with Equalizing Payoffs. However, to the best of our knowledge, no prior literature explicitly describes the dynamics of the EW algorithm to determine which specific bids should be preferred within an energy market framework.

The main objective of this thesis is to characterize the long-term behavior of the players participating in energy auctions depending on their cost functions, capacities, and strategy sets. In Chapter 3, we focus on repeated energy auctions with at least two homogeneous players (i.e., players who share the same capacity and cost function) who have access to only two distinct bids. Within this framework, we derive a set of inequalities that, when satisfied, indicate that all players always prefer the lowest bid. Moreover, by assuming that the cost functions are linear, this set of inequalities is reduced to a single condition. If this condition is not satisfied, we prove two new results. In the former, we show that if the aggregate capacities equal the demand, then almost surely, all players prefer the highest bid. In the latter, we show that if the aggregate capacities exceed the demand, there exists a region in which we switch from a submartingale to a supermartingale regime.

In the second part of the thesis, in Chapter 4, we relax the homogeneous assumption and consider repeated auctions with two players who possess distinct (linear) cost functions, unique capacities, and a choice of at least two distinct bids. In this case, we study four different scenarios based on how the players' strategies overlap. We provide convergence proofs relative to the weights for an arbitrary number of bids in the first two scenarios, and, due to the combinatorial complexity of the last two scenarios, we provide proofs only for two bids.

We begin with the preliminaries to provide the basic definitions and results from the literature in order to understand our work.

2 Preliminaries

In this chapter, we present some definitions and properties utilized throughout the remainder of this thesis. Proofs of these statements are omitted, as they fall outside the scope of this work.

2.1 Game Theory

This section recalls basic game-theoretic definitions from Chapter 7 of Cesa-Bianchi and Lugosi (2006). This background is essential because auctions, in their canonical form, are games, and key results of the Exponential Weights algorithm directly relate to the notion of Nash equilibrium. We begin with the definition of a game.

Definition 1. *A finite normal game G is a set of the form $(\mathcal{K}, (A_k)_{k\in\mathcal{K}}, (u_k)_{k\in\mathcal{K}})$ where $\mathcal{K} = \{1, \dots, K\}$ is the set of players, A_k is the finite set of actions (or pure strategies) of player k and $u_k : \bigotimes_{k=1}^{K} A_k \to \mathbb{R}$ is the payoff function of player k.*

While pure strategies represent deterministic actions, players may also select their actions in a probabilistic manner. To formalize this, let the action set of player k be $A_k = \{1, \dots, N_k\}$.

Definition 2. *A mixed strategy for player k is a probability distribution $\mathbf{p}^{(k)} = (p_1^{(k)}, \dots, p_{N_k}^{(k)})$ over the set A_k.*

If player k uses a mixed strategy, they sample at random one action from A_k according to $\mathbf{p}^{(k)}$. We denote this action by $I^{(k)}$. If all players sample independently, we can define the random vector $\mathbf{I} := (I^{(1)}, \dots, I^{(K)})$ to represent the joint action profile whose realization $\mathbf{i} = (i_1, \dots, i_K)$ is selected with probability:

$$\mathbb{P}[\mathbf{I} = \mathbf{i}] = \mathbb{P}[I^{(1)} = i_1] \cdot \ldots \cdot \mathbb{P}[I^{(K)} = i_K] = p_{i_1}^{(1)} \cdot \ldots \cdot p_{i_K}^{(K)}$$

Under the assumption of independence, the product distribution $\mathbf{p} = \mathbf{p}^{(1)} \times \cdots \times \mathbf{p}^{(K)}$ is also called a mixed strategy profile. Using this notation, the expected payoff of player k

with respect to $\mathbf{p}$ is computed as:

$$\mathbb{E}_{\mathbf{p}}[u_k(\mathbf{I})] = \sum_{\mathbf{i} \in \bigotimes_{j=1}^{K} A_j} \mathbb{P}[\mathbf{I} = \mathbf{i}] \cdot u_k(\mathbf{i}) = \sum_{i_1=1}^{N_1} \cdots \sum_{i_k=1}^{N_K} p_{i_1}^{(1)} \cdot \ldots \cdot p_{i_k}^{(K)} \cdot u_k(\mathbf{i})$$

In principle, we can have infinitely many probability distributions over the set of pure strategies. We denote by $\Delta(A_k)$ the set of all probabilities over A_k and similarly by $\Delta := \Delta(A_1) \times \cdots \times \Delta(A_K)$ the set of all probabilities distributions over the set $\bigotimes_{k=1}^{K} A_k$. Utilizing this notation, we can state the formal definition of Nash equilibrium.

Definition 3. *A mixed strategy profile $\mathbf{p} = \mathbf{p}^{(1)} \times \cdots \times \mathbf{p}^{(K)}$ is called a Nash equilibrium if for all $k = 1, \ldots, K$ and all mixed strategies $\mathbf{q}^{(k)}$, if $\mathbf{p}_k' = \mathbf{p}^{(1)} \times \cdots \times \mathbf{q}^{(k)} \times \cdots \times \mathbf{p}^{(K)}$ denotes the mixed strategy profile obtained by replacing $\mathbf{p}^{(k)}$ by $\mathbf{q}^{(k)}$ and leaving all other players' mixed strategies unchanged, then:*

$$\mathbb{E}_{\mathbf{p}}[u_k] \geq \mathbb{E}_{\mathbf{p}_k'}[u_k]$$

Before discussing the last two definitions, we introduce another notational convention. Assume that an action $\mathbf{i} \in \bigotimes_{k=1}^{K} A_k$ is played. We define $\mathbf{i_{-k}}$ as the $(K-1)$-tuple that contains the actions played by all players except player k. This allows us to introduce two variations of the Nash equilibrium (for reference, see d'Andrea, Gensbittel and Renault (2024)).

Definition 4. *A Nash Equilibrium with Equalizing Payoffs (NEEP) is a Nash equilibrium $\mathbf{p} \in \Delta$ such that for each player k and for all a_k, b_k in the support of $\mathbf{p}^{(k)}$ and all $\mathbf{a_{-k}}$ in the support of $\mathbf{p}_{-k} = \mathbf{p}^{(1)} \times \cdots \times \mathbf{p}^{(k-1)} \times \mathbf{p}^{(k+1)} \times \cdots \times \mathbf{p}^{(K)}$ we have:*

$$u_k(a_k, \mathbf{a}_{-k}) = u_k(b_k, \mathbf{a}_{-k})$$

A NEEP, therefore, describes the equilibrium where all players' realized payoff is independent of the choice of action within the equilibrium support, as every action yields the same payoff against any action profile played by the other players. Note that this definition applies to each player separately; hence, different players may receive different payoffs. Finally, we discuss the notion of Strict Nash equilibrium, which is defined exclusively over pure strategy profiles.

Definition 5. *A Strict Nash Equilibrium of G is an action $\mathbf{i} \in \bigotimes_{k=1}^{K} A_k$ such that for all $k = 1 \ldots, K$ and all $a_k \in A_k$ with $a_k \neq i_k$ we have $u_k(\mathbf{i}) > u_k(a_k, \mathbf{i_{-k}})$.*

2.2 Exponential Weights Algorithm

The Exponential Weights (EW) Algorithm has been widely studied in the literature, and it is a special type of the Hedge(β) algorithm proposed by Freund and Schapire in 1997 (Freund & Schapire, 1997). The inner workings of EW are detailed below in Algorithm 1 (d'Andrea, Gensbittel & Renault, 2024).

Algorithm 1: EW algorithm for player k

Inputs: Number of rounds T, learning rate η_k, initial weights $(w_0(a_k))_{a_k \in A_k}$ such that $w_0(a_k) > 0$ for all $a_k \in A_k$ and $\sum_{a_k \in A_k} w_0(a_k) = 1$.

for $t = 0$ **to** T **do**

Sample action at random $a_k \in A_k$ according to the probability distribution $\mathbf{w}_t$;

Observe action profile $\mathbf{a}^t_{-k}$ of all other players at the end of turn t;

Update weights: For $\forall a_k \in A_k$, compute intermediate weights:

$$\tilde{w}_{t+1}(a_k) = w_t(a_k) \exp\left(\eta_k \cdot u_k(a_k, \mathbf{a}^t_{-k})\right)$$

and normalize to obtain the new weight distribution:

$$w_{t+1}(a_k) = \frac{\tilde{w}_{t+1}(a_k)}{\sum_{b \in A_k} \tilde{w}_{t+1}(b)}.$$

Recent findings by d'Andrea, Gensbittel and Renault (2024) directly related the limits of the EW algorithm to the Nash Equilibrium. Prior to their work, the primary convergence guarantees for the algorithm were in terms of no-regret (Freund & Schapire, 1997) and coarse correlated equilibrium distributions (Hart & Mas-Colell, 2000). The first convergence result relates to the Strict Nash Equilibrium. The authors proved that if such an equilibrium exists, then either this action profile is played exclusively from some finite round onward, or the probability of playing it vanishes entirely.

Theorem 1. *Let* $\mathbf{a}$ *be a strict Nash equilibrium of* G *and* $\mathbf{p}^0 \in \Delta$. *Then* $\mathbb{P}_{\mathbf{p}^0}$*-almost surely:*

$$(\exists t_0 \mid \forall t \geq t_0, \mathbf{a}^t = \mathbf{a}) \text{ or } \left(\mathbf{p}^t(\mathbf{a}) \xrightarrow[t \to \infty]{} 0\right)$$

The second theorem characterizes the geometric structure of the strategy profile's limit when the algorithm is initialized with full support. Let $\text{int}(\Delta) = \{\mathbf{p} \in \Delta \mid p^{(k)}(a_k) > 0 \text{ for all } k \in \mathcal{K}, a_k \in A_k\}$ then we have:

Theorem 2. *For $\mathbf{p}^0 \in int(\Delta)$, any possible limit of the stochastic process $(\mathbf{p}^t)_t$ is a Nash Equilibrium with Equalizing Payoffs.*

While this theorem does not allow us to identify the limit of our process *a priori*, it can help us reinforce the validity of the results we prove in this thesis.

Remark. *Throughout this thesis, the learning rate η of the EW algorithm is constant and equal for all players.*

2.3 Martingales

Due to the inherent stochasticity of the EW algorithm, most of our results rely heavily on martingale theory. Hence, we use this section to recall some important properties. Let $(\mathcal{F}_n)_{n\geq 0}$ be the filtration defined by $\mathcal{F}_n = \sigma(X_0, \dots, X_n)$. These first two definitions are taken from Chapter 14 of Rosenthal (2006).

Definition 6. *A stochastic process $(X)_{n\in\mathbb{N}}$ is a martingale if:*

1. $\mathbb{E}|X_n| < \infty$ *for all n and;*
2. $\mathbb{E}(X_{n+1}|\mathcal{F}_n) = X_n$ *for all n.*

Definition 7. *The sequence $(X)_{n\in\mathbb{N}}$ is a submartingale (supermartingale) if*

1. $\mathbb{E}|X_n| < \infty$ *for all n and;*
2. $\mathbb{E}(X_{n+1}|\mathcal{F}_n) \geq (\leq) X_n$ *for all n.*

Supermartingales and submartingales, provided they satisfy certain boundedness conditions, converge almost surely. This property will prove necessary to discuss the trajectories of the weights assigned to the highest bid in Chapters 3 and 4.

Theorem 3 (**Martingale convergence theorem** (Grimmett & Stirzaker, 2001))**.** *Let $(X)_{n\in\mathbb{N}}$ be a submartingale (or supermartingale) such that $\mathbb{E}|X_n| \leq M$ for some $M > 0$ and all n. Then there exists a random variable X_∞ such that $X_n \xrightarrow{a.s.} X_\infty$ as $n \to \infty$. Moreover if $\mathbb{E}|X_0| < \infty$ then X_∞ has finite mean.*

2.4 Economics of Energy Auctions

In this section, we describe the inner workings of day-ahead energy auctions. The market consists of M energy producers (the bidders/players) and one Power Exchange, which acts as the auctioneer. The duty of the Power Exchange is to procure the demand $Q \in \mathbb{R}^+$ from the producers at minimum cost (Getaneh Abate et al., 2024) (EPEX SPOT, 2025a). Each bidder j possesses a finite set of bids and is characterized by a cost function $C_j(\cdot)$ [1] and a maximum generation capacity denoted by $\overline{x}_j$. In particular, we consider a time horizon of $T \in \mathbb{N}$ rounds during which the cost functions, the available strategy sets, and the total demand remain constant. At each round $t \leq T$, all players submit their bids, and the power exchange determines the optimal allocation by solving the following market-clearing problem:

$$\begin{aligned} \min_{x} \quad & \sum_{j=1}^{M} b_j x_j \\ \text{s.t.} \quad & \sum_{j=1}^{M} x_j = Q \\ & x_j \geq 0 \quad j = 1, \ldots, M \\ & x_j \leq \overline{x}_j \quad j = 1, \ldots, M \end{aligned} \tag{2.1}$$

By solving this problem, the exchange computes the optimal allocation $\mathbf{x}^* = (x_1^*, \ldots, x_M^*)$ and the uniform market clearing price (MCP), λ, which is paid out to every generator whose bid was accepted. In particular, each player j receives a payment equal to λx_j^*. The MCP is the Lagrange multiplier associated with the demand satisfaction constraint. Formally, let $\mathbf{b}, \mathbf{x}, \overline{\mathbf{x}} \in \mathbb{R}^M$, let $\mathbf{1}$ be the vector of ones (of dimension M) and $\mathbf{I}$ the $M \times M$ identity matrix; the Lagrangian function $\mathcal{L}$ is:

$$\mathcal{L}(\mathbf{x}, \lambda, \mu_M, \mu_m) = \mathbf{b}^T\mathbf{x} + \lambda(Q - \mathbf{1}^T\mathbf{x}) + \mu_M^T(\mathbf{I}\mathbf{x} - \overline{\mathbf{x}}) + \mu_m^T(-\mathbf{I}\mathbf{x})$$

[1] Whose specifics are discussed in subsequent chapters.

From stationarity and complementary slackness conditions for player j, we have:

$$b_j - \lambda + \mu_{M,j} - \mu_{m,j} = 0$$

$$\mu_{M,j}(x_j - \overline{x}_j) = 0$$

$$\mu_{m,j} x_j = 0$$

There are three possible scenarios:

1. $x_j = 0$. In this case, the capacity constraint is inactive, and the lower bound is active. This implies that $\mu_{m,j} > 0$ and $\mu_{M,j} = 0$. Therefore $\mu_{m,j} = b_j - \lambda$. This quantity then measures how much player j's bid is overpriced with respect to the MCP.

2. $x_j = \overline{x}_j$. In this case, the capacity constraint is active, and the lower bound is inactive. This implies that $\mu_{m,j} = 0$ and $\mu_{M,j} > 0$. Therefore $\mu_{M,j} = \lambda - b_j$. This quantity then measures how much player j is underbidding with respect to the MCP. However, since we are considering a uniform price auction, in this case, player j does not have an incentive to raise their bid.

3. $0 < x_j < \overline{x}_j$. In this case both are inactive and therefore $\mu_{m,j} = \mu_{M,j} = 0$. This implies that $\lambda = b_j$ and thus this producer is setting the market-clearing price.

Post-auction, exchanges publish the market-clearing data which gives the following information to the players: "(i) hourly market-clearing price, (ii) hourly buy and sell volumes, and (iii) hourly aggregate supply and demand curves" (Getaneh Abate et al., 2024). Each player can then use this information to determine their payoff if they had submitted another bid in their strategy set (and thus use the EW algorithm).

Remark (Tie-Breaking Rule)**.** *If some players submit the same bid and it gets accepted, then they will receive an equal amount of allocation.*

3 Auction with M players and two equal bids

In this chapter, we analyze the behavior of the EW algorithm in repeated auctions with M players who hold two possible bids, b and B, such that $B > b > 0$. We assume that all players share an identical cost function, $C(\cdot)$, which satisfies the following properties:

1. $C(0) = 0$; as producing no energy does not cost anything.
2. $bx - C(x) > 0$ for all $x \in [0, \overline{x}_j]$ and for all $j = 1, \ldots, M$. This is to ensure that the lowest bid never results in a loss for the players.
3. $C(x)$ is an increasing function in x.

Moreover, we impose bounds on the players' capacities (otherwise, there would be too many scenarios to consider). Specifically, we assume that each player's capacity lies in the interval $[\frac{Q}{K}, \frac{Q}{K-1})$ for $K \in \mathbb{N}$ and $M \geq K \geq 2$ [1]. Using these assumptions, in Section 3.1, we derive a set of inequalities that, if satisfied, guarantee that b is always the best bid under any scenario. Then, in Section 3.2, by assuming that the cost function is linear, we: (i) reduce the system of inequalities to just one and (ii) obtain a closed-form expression for the expectation of $w_{t+1}(B)$ conditioned on $w_t(B)$. Next, using this expression for the expectation, we describe the dynamics of the EW algorithm for the case $M = K$. Lastly, for $M > K$, although we cannot describe the results analytically, we provide (with the aid of a simulation) some insights into the expected behavior of the weights.

Remark. *Throughout this chapter, we use the notation $\{b : M - S, B : S\}$ in order to identify the situation in which S players bid B, and the remaining $M - S$ players bid b.*

[1] The lower bound $K \geq 2$ ensures that no single player can satisfy the demand independently. On the other hand, the bound $M \geq K$ ensures market feasibility; if $K > M$, then the aggregate maximum capacity would fail to cover the total demand.

3.1 General setting

In order to derive the set of inequalities, we first study the cases where $K = 2$ and $K = 3$. These allow us to build the necessary intuition for generalizing to an arbitrary K. To find said inequalities, we need to conduct the comparisons between payoffs that the EW algorithm (implicitly) performs; this entails computing all payoffs under all scenarios (i.e., whenever a given number of players bid B rather than b) and then also computing what the payoff would have been if a player had submitted the other bid.

$\underline{K = 2}$

For $K = 2$ the capacity constraints are $\frac{Q}{2} \leq \overline{x}_i < Q$ for all $i = 1, \ldots, M$.

CASE 1 : $\{b : 0, B : M\}$. In this case, all players receive a payoff of $u_i = B\frac{Q}{M} - C\left(\frac{Q}{M}\right)$. If any player j had bid b instead, their payoff would have been $\hat{u}_j = B\overline{x}_j - C(\overline{x}_j)$ [2].

In the following cases, we split the page into two columns. On the left side, we study the auction from the point of view of the players that bid b, while on the right side, we study it from the perspective of the players that bid B.

CASE 2 = $\{b : 1, B : M - 1\}$.

The player who bid b receives a payoff of:

$$u_j = B\overline{x}_j - C(\overline{x}_j)$$

If player j had bid B instead they would have gotten:

$$\hat{u}_j = B\frac{Q}{M} - C\left(\frac{Q}{M}\right)$$

The players who bid B receive a payoff of:

$$u_i = B\frac{Q - \overline{x}_j}{M - 1} - C\left(\frac{Q - \overline{x}_j}{M - 1}\right)$$

for all $i \neq j$. If player i had bid b instead then they would have gotten [3]:

$$\hat{u}_i = b\frac{Q}{2} - C\left(\frac{Q}{2}\right)$$

[2] Recall that we are in a uniform price setting. Since the demand cannot be satisfied by player j alone, the exchange accepts the bids of the players who bid B as well. Therefore, the market clearing price is B.

[3] In this case the clearing price is b because the two players that bid b can satisfy the full demand by themselves.

CASE 3 = $\{b : 2, B : M-2\}$. Let S_b denote the set of indices of the players that bid b.

The players who bid b receive:

$$u_j = b\frac{Q}{2} - C\left(\frac{Q}{2}\right)$$

for all $j \in S_b$. If player j had bid B instead they would have gotten:

$$\hat{u}_j = B\frac{Q-\overline{x}_k}{M-1} - C\left(\frac{Q-\overline{x}_k}{M-1}\right)$$

where $k = S_b - \{j\}$.

The players who bid B receive:

$$u_i = 0$$

for all $i \notin S_b$. If player i had bid b instead then they would have gotten:

$$\hat{u}_i = b\frac{Q}{3} - C\left(\frac{Q}{3}\right)$$

OTHER CASES: In the remaining $M-2$ cases, at least three players bid b; these players alone provide enough capacity to satisfy the demand. Therefore, none of the players who bid b has an incentive to switch bids, as such a player would be excluded from the allocation and receive a zero payoff. Therefore, from the point of view of player i, the weights associated with bid b in the EW algorithm will monotonically increase whenever these inequalities hold.

$$B\overline{x}_i - C(\overline{x}_i) > B\frac{Q}{M} - C\left(\frac{Q}{M}\right)$$

$$b\frac{Q}{2} - C\left(\frac{Q}{2}\right) > B\frac{Q-\overline{x}_j}{M-1} - C\left(\frac{Q-\overline{x}_j}{M-1}\right) \quad \text{for all } j = 1\ldots, M \text{ with } i \neq j$$

Next, we derive the inequalities for $K = 3$.

<u>$K = 3$</u>

In this case, the capacity constraints are $\frac{Q}{3} \leq \overline{x}_i < \frac{Q}{2}$ for all $i = 1, \ldots, M$.

CASE 1: $\{b : 0, B : M\}$. The payoffs are the same as for CASE 1 in $K = 2$.

CASE 2: $\{b : 1, B : M-1\}$.

The players who bid b receive:

$$u_j = B\overline{x}_j - C(\overline{x}_j)$$

If player j had bid B instead then they would have gotten:

$$\hat{u}_j = B\frac{Q}{M} - C\left(\frac{Q}{M}\right)$$

The players who bid B receive:

$$u_i = B\frac{Q-\overline{x}_j}{M-1} - C\left(\frac{Q-\overline{x}_j}{M-1}\right)$$

for all $i \neq j$. If player i had bid b instead then they would have gotten:

$$\hat{u}_i = B\overline{x}_i - C(\overline{x}_i)$$

CASE 3: $\{b : 2, B : M-2\}$

The players who bid b receive:

$$u_j = B\overline{x}_j - C(\overline{x}_j)$$

for all $j \in S_b$. If player j had bid B instead then they would have gotten:

$$\hat{u}_j = B\frac{Q-\overline{x}_k}{M-1} - C\left(\frac{Q-\overline{x}_k}{M-1}\right)$$

where $k = S_b - \{j\}$.

The players who bid B receive:

$$u_i = B\frac{Q-\sum\limits_{j\in S_b}\overline{x}_j}{M-2} - C\left(\frac{Q-\sum\limits_{j\in S_b}\overline{x}_j}{M-2}\right)$$

for all $i \notin S_b$. If player i had bid b instead then they would have gotten:

$$\hat{u}_i = b\frac{Q}{3} - C(\frac{Q}{3})$$

CASE 4: $\{b : 3, B : M-3\}$

The players who bid b receive:

$$u_j = b\frac{Q}{3} - C(\frac{Q}{3})$$

for all $j \in S_b$. If player j had bid B instead then they would have gotten:

$$\hat{u}_j = B\frac{Q-\sum\limits_{\substack{k\in S_b\\k\neq j}}\overline{x}_k}{M-2} - C\left(\frac{Q-\sum\limits_{\substack{k\in S_b\\k\neq j}}\overline{x}_k}{M-2}\right)$$

The players who bid B receive:

$$u_i = 0$$

for all $i \notin S_b$. If player i had bid b instead then they would have gotten:

$$\hat{u}_i = b\frac{Q}{4} - C(\frac{Q}{4})$$

Therefore, analogously to $K = 2$, from the perspective of player i, the weight associated with b in the EW algorithm will monotonically increase whenever these inequalities hold:

$$B\overline{x}_i - C(\overline{x}_i) > B\frac{Q}{M} - C\left(\frac{Q}{M}\right)$$

$$B\overline{x}_i - C(\overline{x}_i) > B\frac{Q-\overline{x}_j}{M-1} - C\left(\frac{Q-\overline{x}_j}{M-1}\right) \quad \text{for all } j = 1\ldots, M \text{ with } i \neq j$$

$$b\frac{Q}{3} - C\left(\frac{Q}{3}\right) > B\frac{Q-\sum\limits_{j\in S_b}\overline{x}_j}{M-2} - C\left(\frac{Q-\sum\limits_{j\in S_b}\overline{x}_j}{M-2}\right) \quad \text{for all } S_b \text{ with } |S_b| = 2 \text{ and } i \notin S_b.$$

<u>K</u>

From $K = 2$ and $K = 3$, we can clearly notice 3 patterns:

1. On the left-hand side, the Kth term is of the form $b\frac{Q}{K} - C(\frac{Q}{K})$.
2. The remaining $K-1$ equations on the left-hand side are all of the form $B\overline{x}_i - C(\overline{x}_i)$.
3. The expressions on the right-hand side can all be written as:

$$B\frac{Q-\sum\limits_{j\in S_b}\overline{x}_j}{M-S} - C(\frac{Q-\sum\limits_{j\in S_b}\overline{x}_j}{M-S})$$

   where $|S_b| = S$. The K terms are then obtained by starting with $S = 0$ and proceeding up to $S = K-1$.

Therefore, in the case where the capacities are bounded by $\frac{Q}{K} \leq \overline{x}_i < \frac{Q}{K-1}$ for all $i = 1, \ldots, M$, the weight of b monotonically increases (from the perspective of player i) when these inequalities are satisfied:

$$
\begin{aligned}
&B\overline{x}_i - C(\overline{x}_i) > B\frac{Q}{M} - C\left(\frac{Q}{M}\right) \\
&B\overline{x}_i - C(\overline{x}_i) > B\frac{Q-\overline{x}_j}{M-1} - C\left(\frac{Q-\overline{x}_j}{M-1}\right) \quad \text{for all } j = 1\ldots, M \text{ with } i \neq j \\
&B\overline{x}_i - C(\overline{x}_i) > B\frac{Q-\sum\limits_{j\in S_b}\overline{x}_j}{M-2} - C\left(\frac{Q-\sum\limits_{j\in S_b}\overline{x}_j}{M-2}\right) \quad \text{for all } S_b \text{ with } |S_b| = 2 \text{ and } i \notin S_b. \\
&\qquad\vdots \\
&B\overline{x}_i - C(\overline{x}_i) > B\frac{Q-\sum\limits_{j\in S_b}\overline{x}_j}{M-(K-2)} - C\left(\frac{Q-\sum\limits_{j\in S_b}\overline{x}_j}{M-(K-2)}\right) \quad \text{for all } S_b \text{ with } |S_b| = K-2 \text{ and } i \notin S_b. \\
&b\frac{Q}{K} - C\left(\frac{Q}{K}\right) > B\frac{Q-\sum\limits_{j\in S_b}\overline{x}_j}{M-(K-1)} - C\left(\frac{Q-\sum\limits_{j\in S_b}\overline{x}_j}{M-(K-1)}\right) \quad \text{for all } S_b \text{ with } |S_b| = K-1 \text{ and } i \notin S_b.
\end{aligned}
\tag{3.1}
$$

From the set of inequalities (3.1), we can see that, as M increases, the right-hand side of the inequalities approaches 0. Therefore, if we fix all variables except M, there exists a $\hat{M}$ above which the inequalities are all satisfied. Although we know this $\hat{M}$ does exist, it is still computationally expensive to verify whether these inequalities are satisfied because there are $\binom{M}{S}$ distinct subsets of size S we need to consider for each inequality. However, what if the inequalities are not satisfied? What are the dynamics of the weights then?

In the next section, we assume that the cost functions are linear and that all players have the same capacity. This simplification allows us to reduce the number of inequalities to just one. It also allows us to obtain an expression for the conditional expectation $\mathbb{E}[w_{t+1}(B)|w_t(B)]$ that we use to study the dynamics when the inequality is not satisfied.

3.2 Equal capacity with Linear costs

As previously mentioned, in this section, we make the following additional assumptions:

1. Each player shares the same capacity. That is $\overline{x}_i = \overline{x}$ for all $i = 1, \ldots, M$.

2. The cost function of the players is $C(x) = ax$ for some $a > 0$. Moreover, in order to satisfy the properties of the cost function outlined at the beginning of Chapter 3, we must have $b > a$.

Note the first $K-1$ terms on the left-hand side of the system of inequalities (3.1) are all equal. This means that if we can identify a maximum among their right-hand-side counterparts, we can reduce the first $K-1$ inequalities to one. Using the same-capacity assumption, the quantities of energy on the right-hand side can be written as:

$$\frac{Q - \sum\limits_{j \in S_b} \overline{x}_j}{M - S} = \frac{Q - S\overline{x}}{M - S} := q(S)$$

for $S = 0, \ldots, K-2$ and $|S_b| = S$. Using this definition, we show that as more players are able to deliver at their full capacity (i.e., they bid b), the quantity to be split among the remaining players decreases. That is:

$$q(S) - q(S+1) = \frac{Q - S\overline{x}}{M - S} - \frac{Q - (S+1)\overline{x}}{M - (S+1)}$$

$$> (Q - S\overline{x})(M - S - 1) + (Q - (S+1)\overline{x})(M - S) = -Q + M\overline{x} > 0$$

where the last inequality follows from the fact that $M \geq K$ and $\overline{x} \geq Q/K$. Combining this result with the linearity of the cost function allows us to determine that the terms on the right-hand side are decreasing from top to bottom:

$$Bq(S) - aq(S) - [Bq(S+1) - aq(S+1)] = (B-a)[q(S) - q(S+1)] > 0$$

This shows that the highest payoff is achieved for $S = 0$ and thus, the inequalities to check are reduced to:

$$B\overline{x} - a\overline{x} > B\frac{Q}{M} - a\frac{Q}{M}$$

$$b\frac{Q}{K} - a\frac{Q}{K} > B\frac{Q - (K-1)\overline{x}}{M - (K-1)} - a\frac{Q - (K-1)\overline{x}}{M - (K-1)}$$

Note that the first inequality is not satisfied for $M = K$ and $\overline{x} = \frac{Q}{K}$ as we have equality. However, in this case, the weights would simply remain constant. In all other cases, we have a strict inequality. Therefore, the only inequality we need to check to determine if $w_t(b)$ is

increasing is:

$$(b-a)\frac{Q}{K} > (B-a)\frac{Q-(K-1)\overline{x}}{M-(K-1)} \tag{3.2}$$

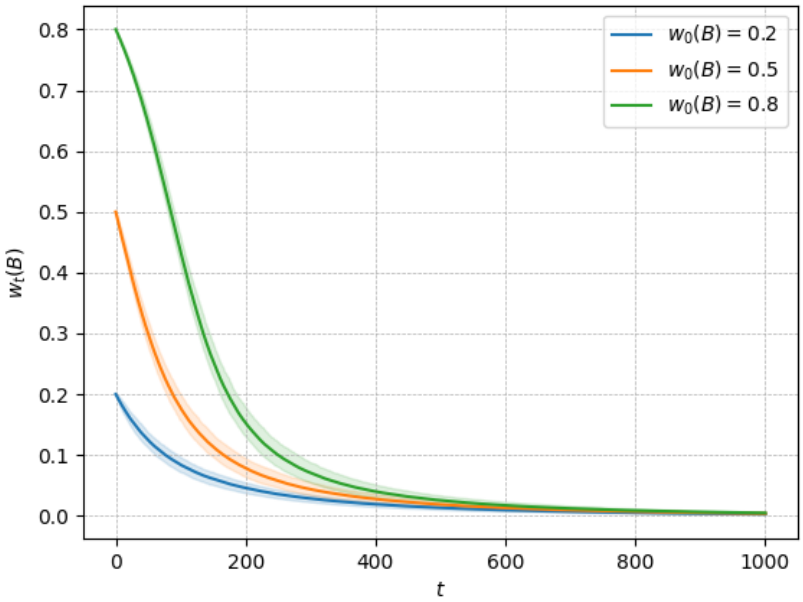


Figure 3.1: Plot of $w_t(B)$ over time for different starting values for an auction with the following characteristics: $Q = 11.484$, $T = 1000$, $(M, K) = (4, 4)$, $\overline{x} = 3.6 > \frac{Q}{K}$, $(B, b) = (1.1, 0.8)$ and $a = 0.7$. The solid line represents the mean, while the shaded areas cover the 90% confidence interval over 500 runs.

Figure 3.1 [4] illustrates the case in which Inequality (3.2) is satisfied. Since bid b is preferred in all scenarios (except one where bidding b or B gives the same payoff), the weight of bid B decreases over time, regardless of its initial value. Another noteworthy observation is the narrow confidence intervals around the mean, which show that in this scenario, the dynamic is predictable with high probability.

In what follows, we assume that Inequality (3.2) is not satisfied. In this case, it is not immediately obvious what the dynamics should be. Therefore, we begin by finding the conditional expectation with respect to the weights associated with B, which can be written

[4] The model was implemented in Python 3.12 using the Pyomo (Hart, Watson & Woodruff, 2011; Bynum et al., 2021). All the tests were executed using an Intel Core i7-12700H 4.7GHz and 16GB RAM. Code available at https://github.com/scinii/repeated-energy-auctions.

in the following manner:

$$\mathbb{E}[w_{t+1}(B)|w_t(B)] = \sum_{S=0}^{M} \mathbb{P}(\text{S bidders bid } B) \cdot \frac{w_t(B)\exp[\eta u(x_{B,t})]}{w_t(B)\exp[\eta u(x_{B,t})] + w_t(b)\exp[\eta u(x_{b,t})]}$$

where $u(x_{B,t})$ and $u(x_{b,t})$ are the payoffs received by the player if he had bid B or b at time t (while keeping fixed all the other players' bids). Note that this expectation is the same for all players, as they all share identical characteristics and their weights evolve in the same manner. However, to evaluate the individual terms inside the sum, we need to consider six different cases separately. This is necessary as this is an auction with one clearing price, and therefore the specific payout will be different if our reference player (let's call this player P) bids b or B.

1. $E_1 = \{b : 0, B : M\}$. In this case, we have:

$$\mathbb{P}(E_1) \cdot \frac{w_t(B)\exp[\eta(B-a)\frac{Q}{M}]}{w_t(B)\exp[\eta(B-a)\frac{Q}{M}] + w_t(b)\exp[\eta(B-a)\overline{x}]}$$

$$= w_t(B)^M \cdot \frac{w_t(B)\exp[\eta(B-a)\frac{Q}{M}]}{w_t(B)\exp[\eta(B-a)\frac{Q}{M}] + w_t(b)\exp[\eta(B-a)\overline{x}]}$$

2. $E_2 = \{b : M-S, B : S\}$ for $S = M-(K-2), \ldots, M-1$. In this case, we must consider two possibilities:

   a) If $P \in S_b$ (i.e. if our reference player P bid b) then:

$$\mathbb{P}(E_2 \wedge P \in S_b) \cdot \frac{w_t(B)\exp[\eta(B-a)\frac{Q-(M-S-1)\overline{x}}{S+1}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(M-S-1)\overline{x}}{S+1}] + w_t(b)\exp[\eta(B-a)\overline{x}]}$$

$$= \binom{M-1}{M-S-1} w_t(b)^{M-S} w_t(B)^S \frac{w_t(B)\exp[\eta(B-a)\frac{Q-(M-S-1)\overline{x}}{S+1}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(M-S-1)\overline{x}}{S+1}] + w_t(b)\exp[\eta(B-a)\overline{x}]}$$

   b) If $P \notin S_b$ then:

$$\mathbb{P}(E_2 \wedge P \notin S_b) \cdot \frac{w_t(B)\exp[\eta(B-a)\frac{Q-(M-S)\overline{x}}{S}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(M-S)\overline{x}}{S}] + w_t(b)\exp[\eta(B-a)\overline{x}]}$$

$$= \binom{M-1}{M-S} w_t(b)^{M-S} w_t(B)^S \frac{w_t(B)\exp[\eta(B-a)\frac{Q-(M-S)\overline{x}}{S}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(M-S)\overline{x}}{S}] + w_t(b)\exp[\eta(B-a)\overline{x}]}$$

3. $E_3 = \{b : K-1, B : M-(K-1)\}$. Again, we must consider two possibilities:

   a) If $P \in S_b$ then:

$$\mathbb{P}(E_3 \wedge P \in S_b) \cdot \frac{w_t(B)\exp[\eta(B-a)\frac{Q-(K-2)\overline{x}}{M-(K-2)}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(K-2)\overline{x}}{M-(K-2)}] + w_t(b)\exp[\eta(B-a)\overline{x}]}$$

$$= \binom{M-1}{K-2} w_t(b)^{K-1} w_t(B)^{M-(K-1)} \frac{w_t(B)\exp[\eta(B-a)\frac{Q-(K-2)\overline{x}}{M-(K-2)}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(K-2)\overline{x}}{M-(K-2)}] + w_t(b)\exp[\eta(B-a)\overline{x}]}$$

   b) If $P \notin S_b$ then:

$$\mathbb{P}(E_3 \wedge P \notin S_b) \cdot \frac{w_t(B)\exp[\eta(B-a)\frac{Q-(K-1)\overline{x}}{M-(K-1)}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(K-1)\overline{x}}{M-(K-1)}] + w_t(b)\exp[\eta(b-a)\frac{Q}{K}]}$$

$$= \binom{M-1}{K-1} w_t(b)^{K-1} w_t(B)^{M-(K-1)} \frac{w_t(B)\exp[\eta(B-a)\frac{Q-(K-1)\overline{x}}{M-(K-1)}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(K-1)\overline{x}}{M-(K-1)}] + w_t(b)\exp[\eta(b-a)\frac{Q}{K}]}$$

   Note that the reason why E_3 could not be included in E_2 is that when $P \notin S_b$, the payoff associated with bid b is different.

4. $E_4 = \{b : K, B : M-K\}$. Again, we consider two cases:

   a) If $P \in S_b$ then:

$$\mathbb{P}(E_4 \wedge P \in S_b) \cdot \frac{w_t(B)\exp[\eta(B-a)\frac{Q-(K-1)\overline{x}}{M-(K-1)}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(K-1)\overline{x}}{M-(K-1)}] + w_t(b)\exp[\eta(b-a)\frac{Q}{K}]}$$

$$= \binom{M-1}{K-1} w_t(b)^{K} w_t(B)^{M-K} \frac{w_t(B)\exp[\eta(B-a)\frac{Q-(K-1)\overline{x}}{M-(K-1)}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(K-1)\overline{x}}{M-(K-1)}] + w_t(b)\exp[\eta(b-a)\frac{Q}{K}]}$$

b) If $P \notin S_b$ then:

$$\mathbb{P}(E_3 \wedge P \notin S_b) \cdot \frac{w_t(B)\exp[\eta(B-a)0]}{w_t(B)\exp[\eta(B-a)0] + w_t(b)\exp[\eta(b-a)\frac{Q}{K+1}]}$$

$$= \binom{M-1}{K} w_t(b)^K w_t(B)^{M-K} \frac{w_t(B)}{w_t(B) + w_t(b)\exp[\eta(b-a)\frac{Q}{K+1}]}$$

5. $E_5 = \{b : M-S, B : S\}$ for $S = 1, \ldots, M-K-1$.

a) If $P \in S_b$ then:

$$\mathbb{P}(E_5 \wedge P \in S_b) \cdot \frac{w_t(B)\exp[\eta(B-a)0]}{w_t(B)\exp[\eta(B-a)0] + w_t(b)\exp[\eta(b-a)\frac{Q}{M-S}]}$$

$$= \binom{M-1}{M-S-1} w_t(b)^{M-S} w_t(B)^S \frac{w_t(B)}{w_t(B) + w_t(b)\exp[\eta(b-a)\frac{Q}{M-S}]}$$

b) If $P \notin S_b$ then:

$$\mathbb{P}(E_5 \wedge P \notin S_b) \cdot \frac{w_t(B)\exp[\eta(B-a)0]}{w_t(B)\exp[\eta(B-a)0] + w_t(b)\exp[\eta(b-a)\frac{Q}{M-S+1}]}$$

$$= \binom{M-1}{M-S} w_t(b)^{M-S} w_t(B)^S \frac{w_t(B)}{w_t(B) + w_t(b)\exp[\eta(b-a)\frac{Q}{M-S+1}]}$$

6. $E_6 = \{b : M, B : 0\}$

$$\mathbb{P}(E_6) \cdot \frac{w_t(B)\exp[\eta(B-a)0]}{w_t(B)\exp[\eta(B-a)0] + w_t(b)\exp[\eta(B-a)\frac{Q}{M}]}$$

$$= w_t(b)^M \frac{w_t(B)}{w_t(B) + w_t(b)\exp[\eta(b-a)\frac{Q}{M}]}$$

Therefore, combining all these cases together, the expectation is:

$$\mathbb{E}[w_{t+1}(B)|w_t(B)] = w_t(B)^M \cdot \frac{w_t(B)\exp[\eta(B-a)\frac{Q}{M}]}{w_t(B)\exp[\eta(B-a)\frac{Q}{M}] + w_t(b)\exp[\eta(B-a)\overline{x}]}$$

$$+ \sum_{S=M-(K-2)}^{M-1} \binom{M-1}{M-S-1} w_t(b)^{M-S} w_t(B)^S \frac{w_t(B)\exp[\eta(B-a)\frac{Q-(M-S-1)\overline{x}}{S+1}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(M-S-1)\overline{x}}{S+1}] + w_t(b)\exp[\eta(B-a)\overline{x}]}$$

$$+ \sum_{S=M-(K-2)}^{M-1} \binom{M-1}{M-S} w_t(b)^{M-S} w_t(B)^S \frac{w_t(B)\exp[\eta(B-a)\frac{Q-(M-S)\overline{x}}{S}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(M-S)\overline{x}}{S}] + w_t(b)\exp[\eta(B-a)\overline{x}]}$$

$$+ \binom{M-1}{K-2} w_t(b)^{K-1} w_t(B)^{M-(K-1)} \frac{w_t(B)\exp[\eta(B-a)\frac{Q-(K-2)\overline{x}}{M-(K-2)}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(K-2)\overline{x}}{M-(K-2)}] + w_t(b)\exp[\eta(B-a)\overline{x}]}$$

$$+ \binom{M-1}{K-1} w_t(b)^{K-1} w_t(B)^{M-(K-1)} \frac{w_t(B)\exp[\eta(B-a)\frac{Q-(K-1)\overline{x}}{M-(K-1)}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(K-1)\overline{x}}{M-(K-1)}] + w_t(b)\exp[\eta(b-a)\frac{Q}{K}]}$$

$$+ \binom{M-1}{K-1} w_t(b)^{K} w_t(B)^{M-K} \frac{w_t(B)\exp[\eta(B-a)\frac{Q-(K-1)\overline{x}}{M-(K-1)}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(K-1)\overline{x}}{M-(K-1)}] + w_t(b)\exp[\eta(b-a)\frac{Q}{K}]}$$

$$+ \binom{M-1}{K} w_t(b)^{K} w_t(B)^{M-K} \frac{w_t(B)}{w_t(B) + w_t(b)\exp[\eta(b-a)\frac{Q}{K+1}]}$$

$$+ \sum_{S=1}^{M-K-1} \binom{M-1}{M-S-1} w_t(b)^{M-S} w_t(B)^S \frac{w_t(B)}{w_t(B) + w_t(b)\exp[\eta(b-a)\frac{Q}{M-S}]}$$

$$+ \sum_{S=1}^{M-K-1} \binom{M-1}{M-S} w_t(b)^{M-S} w_t(B)^S \frac{w_t(B)}{w_t(B) + w_t(b)\exp[\eta(b-a)\frac{Q}{M-S+1}]}$$

$$+ w_t(b)^M \frac{w_t(B)}{w_t(B) + w_t(b)\exp[\eta(b-a)\frac{Q}{M}]}$$

This expression can be simplified. Firstly, consider the fifth and sixth term:

$$\binom{M-1}{K-1} w_t(b)^{K-1} w_t(B)^{M-(K-1)} \frac{w_t(B)\exp[\eta(B-a)\frac{Q-(K-1)\overline{x}}{M-(K-1)}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(K-1)\overline{x}}{M-(K-1)}] + w_t(b)\exp[\eta(b-a)\frac{Q}{K}]}$$

$$+ \binom{M-1}{K-1} w_t(b)^{K} w_t(B)^{M-K} \frac{w_t(B)\exp[\eta(B-a)\frac{Q-(K-1)\overline{x}}{M-(K-1)}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(K-1)\overline{x}}{M-(K-1)}] + w_t(b)\exp[\eta(b-a)\frac{Q}{K}]}$$

$$= w_t(B)\binom{M-1}{K-1} w_t(b)^{K-1} w_t(B)^{M-K} \frac{w_t(B)\exp[\eta(B-a)\frac{Q-(K-1)\overline{x}}{M-(K-1)}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(K-1)\overline{x}}{M-(K-1)}] + w_t(b)\exp[\eta(b-a)\frac{Q}{K}]}$$

$$+ w_t(b)\binom{M-1}{K-1} w_t(b)^{K-1} w_t(B)^{M-K} \frac{w_t(B)\exp[\eta(B-a)\frac{Q-(K-1)\overline{x}}{M-(K-1)}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(K-1)\overline{x}}{M-(K-1)}] + w_t(b)\exp[\eta(b-a)\frac{Q}{K}]}$$

$$= \binom{M-1}{K-1} w_t(b)^{K-1} w_t(B)^{M-K} \frac{w_t(B)\exp[\eta(B-a)\frac{Q-(K-1)\overline{x}}{M-(K-1)}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(K-1)\overline{x}}{M-(K-1)}] + w_t(b)\exp[\eta(b-a)\frac{Q}{K}]}$$

where the last equality comes from the fact that the weights sum to 1. Next, let us consider the first four terms. The first can be included in the second summation (third term) via $S = M$, while the fourth term can be included in the first summation (second term) via $S = M - (K-1)$.

$$\sum_{S=M-(K-1)}^{M-1} \binom{M-1}{M-S-1} w_t(b)^{M-S} w_t(B)^{S} \frac{w_t(B)\exp[\eta(B-a)\frac{Q-(M-S-1)\overline{x}}{S+1}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(M-S-1)\overline{x}}{S+1}] + w_t(b)\exp[\eta(B-a)\overline{x}]}$$

$$+ \sum_{S=M-(K-2)}^{M} \binom{M-1}{M-S} w_t(b)^{M-S} w_t(B)^{S} \frac{w_t(B)\exp[\eta(B-a)\frac{Q-(M-S)\overline{x}}{S}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(M-S)\overline{x}}{S}] + w_t(b)\exp[\eta(B-a)\overline{x}]}$$

Furthermore, we can shift the second sum by 1 and obtain:

$$\sum_{S=M-(K-1)}^{M-1}\binom{M-1}{M-S-1}w_t(b)^{M-S}w_t(B)^S\frac{w_t(B)\exp[\eta(B-a)\frac{Q-(M-S-1)\overline{x}}{S+1}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(M-S-1)\overline{x}}{S+1}]+w_t(b)\exp[\eta(B-a)\overline{x}]}$$

$$+\sum_{S=M-(K-1)}^{M-1}\binom{M-1}{M-S-1}w_t(b)^{M-S-1}w_t(B)^{S+1}\frac{w_t(B)\exp[\eta(B-a)\frac{Q-(M-S-1)\overline{x}}{S+1}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(M-S-1)\overline{x}}{S+1}]+w_t(b)\exp[\eta(B-a)\overline{x}]}$$

$$=w_t(b)\sum_{S=M-(K-1)}^{M-1}\binom{M-1}{M-S-1}w_t(b)^{M-S-1}w_t(B)^S\frac{w_t(B)\exp[\eta(B-a)\frac{Q-(M-S-1)\overline{x}}{S+1}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(M-S-1)\overline{x}}{S+1}]+w_t(b)\exp[\eta(B-a)\overline{x}]}$$

$$+w_t(B)\sum_{S=M-(K-1)}^{M-1}\binom{M-1}{M-S-1}w_t(b)^{M-S-1}w_t(B)^S\frac{w_t(B)\exp[\eta(B-a)\frac{Q-(M-S-1)\overline{x}}{S+1}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(M-S-1)\overline{x}}{S+1}]+w_t(b)\exp[\eta(B-a)\overline{x}]}$$

$$=\sum_{S=M-(K-1)}^{M-1}\binom{M-1}{M-S-1}w_t(b)^{M-S-1}w_t(B)^S\frac{w_t(B)\exp[\eta(B-a)\frac{Q-(M-S-1)\overline{x}}{S+1}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(M-S-1)\overline{x}}{S+1}]+w_t(b)\exp[\eta(B-a)\overline{x}]}$$

The last step to simplify the expectation is to combine the last four terms analogously to the first four terms. The seventh term can be placed inside the second summation (ninth term) via $S=M-K$, and the tenth term can be placed inside the first sum (eighth term) via $S=0$.

$$\sum_{S=0}^{M-K-1}\binom{M-1}{M-S-1}w_t(b)^{M-S}w_t(B)^S\frac{w_t(B)}{w_t(B)+w_t(b)\exp[\eta(b-a)\frac{Q}{M-S}]}$$

$$+\sum_{S=1}^{M-K}\binom{M-1}{M-S}w_t(b)^{M-S}w_t(B)^S\frac{w_t(B)}{w_t(B)+w_t(b)\exp[\eta(b-a)\frac{Q}{M-S+1}]}$$

$$\sum_{S=0}^{M-K-1}\binom{M-1}{M-S-1}w_t(b)^{M-S}w_t(B)^S\frac{w_t(B)}{w_t(B)+w_t(b)\exp[\eta(b-a)\frac{Q}{M-S}]}$$

$$+\sum_{S=0}^{M-K-1}\binom{M-1}{M-S-1}w_t(b)^{M-S-1}w_t(B)^{S+1}\frac{w_t(B)}{w_t(B)+w_t(b)\exp[\eta(b-a)\frac{Q}{M-S}]}$$

$$=\sum_{S=0}^{M-K-1}\binom{M-1}{M-S-1}w_t(b)^{M-S-1}w_t(B)^S\frac{w_t(B)}{w_t(B)+w_t(b)\exp[\eta(b-a)\frac{Q}{M-S}]}$$

Therefore, the expectation can be written as:

$$\mathbb{E}[w_{t+1}(B)|\mathcal{F}_t] =$$

$$\sum_{S=M-(K-1)}^{M-1} \binom{M-1}{M-S-1} w_t(b)^{M-S-1} w_t(B)^S \frac{w_t(B)\exp\left[\eta(B-a)\frac{Q-(M-S-1)\overline{x}}{S+1}\right]}{w_t(B)\exp\left[\eta(B-a)\frac{Q-(M-S-1)\overline{x}}{S+1}\right] + w_t(b)\exp\left[\eta(B-a)\overline{x}\right]}$$

$$+ \binom{M-1}{K-1} w_t(b)^{K-1} w_t(B)^{M-K} \frac{w_t(B)\exp\left[\eta(B-a)\frac{Q-(K-1)\overline{x}}{M-(K-1)}\right]}{w_t(B)\exp\left[\eta(B-a)\frac{Q-(K-1)\overline{x}}{M-(K-1)}\right] + w_t(b)\exp\left[\eta(b-a)\frac{Q}{K}\right]}$$

$$+ \sum_{S=0}^{M-K-1} \binom{M-1}{M-S-1} w_t(b)^{M-S-1} w_t(B)^S \frac{w_t(B)}{w_t(B) + w_t(b)\exp\left[\eta(b-a)\frac{Q}{M-S}\right]}$$

As mentioned in the introduction of this chapter, analytical findings could only be obtained for $M = K$. Under this condition, the expectation is:

$$\mathbb{E}[w_{t+1}(B)|w_t(B)] =$$

$$\sum_{S=1}^{K-1} \binom{K-1}{S} w_t(b)^{K-S-1} w_t(B)^S \frac{w_t(B)\exp[\eta(B-a)\frac{Q-(K-S-1)\overline{x}}{S+1}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(K-S-1)\overline{x}}{S+1}] + w_t(b)\exp[\eta(B-a)\overline{x}]}$$

$$+ w_t(b)^{K-1} \frac{w_t(B)\exp[\eta(B-a)(Q-(K-1)\overline{x})]}{w_t(B)\exp[\eta(B-a)(Q-(K-1)\overline{x})] + w_t(b)\exp[\eta(b-a)\frac{Q}{K}]} \tag{3.3}$$

In the following result, we demonstrate that if the players' aggregate capacities equal the demand, then $w_t(B)$ converges to 1 almost surely.

Theorem 4. *If Inequality (3.2) does not hold, $\overline{x} = \frac{Q}{K}$, and $M = K$, then for any non-trivial initial weight $w_0(B) \in (0,1)$ we have $w_t(B) \xrightarrow{a.s.} 1$.*

Proof. Given that $M = K$, we consider the expectation derived in Equation (3.3). Substi-

tuting $\overline{x} = \frac{Q}{K}$ yields:

$$\begin{aligned}
\mathbb{E}[w_{t+1}(B)|w_t(B)] = \\
&+ \sum_{S=1}^{K-1} \binom{K-1}{S} w_t(b)^{K-S-1} w_t(B)^S \frac{w_t(B)\exp[\eta(B-a)\frac{Q}{K}]}{w_t(B)\exp[\eta(B-a)\frac{Q}{K}] + w_t(b)\exp[\eta(B-a)\frac{Q}{K}]} \\
&+ w_t(b)^{K-1} \frac{w_t(B)\exp[\eta(B-a)\frac{Q}{K}]}{w_t(B)\exp[\eta(B-a)\frac{Q}{K}] + w_t(b)\exp[\eta(b-a)\frac{Q}{K}]} \\
&= \sum_{S=1}^{K-1} \binom{K-1}{S} w_t(b)^{K-S-1} w_t(B)^S w_t(B) + w_t(b)^{K-1} \frac{w_t(B)}{w_t(B) + w_t(b)\exp[\eta(b-B)\frac{Q}{K}]} \\
&= w_t(B)(1 - w_t(b)^{k-1}) + w_t(b)^{K-1} \frac{w_t(B)}{w_t(B) + w_t(b)\exp[\eta(b-B)\frac{Q}{K}]}
\end{aligned}$$

where the last equality follows from the binomial theorem. Therefore, the conditional expectation can be rewritten as:

$$\begin{aligned}
\mathbb{E}[w_{t+1}(B)|w_t(B)] &= w_t(B)(1 - w_t(b)^{k-1}) + w_t(b)^{K-1} \frac{w_t(B)}{w_t(B) + w_t(b)\exp[\eta(b-B)\frac{Q}{K}]} \\
&= w_t(B)\left[1 + w_t(b)^{K-1}\left(\frac{1}{w_t(B) + w_t(b)\exp[\eta(b-B)\frac{Q}{K}]} - 1\right)\right] = w_t(B)\rho_t
\end{aligned}$$

where ρ_t denotes the expression within the outer brackets. Because $b - B < 0$ it follows that $\rho_t \geq 1$ with equality holding if and only if $w_t(B) \in \{0, 1\}$. Since we assumed a non-trivial initialization, the weight $w_t(B)$ cannot reach the boundary values $\{0, 1\}$ for any finite t. Thus we have $\rho_t > 1$ for all t which implies that:

$$\mathbb{E}[w_{t+1}(B)|w_t(B)] > w_t(B) \tag{3.4}$$

Therefore, by Theorem 3, the sequence $w_t(B)$ converges almost surely to a random variable $w_\infty(B)$ with finite mean. Moreover, by applying the law of total expectation to Inequality (3.4) we obtain:

$$\mathbb{E}[w_{t+1}(B)] > \mathbb{E}[w_t(B)] \tag{3.5}$$

Since $E[w_t(B)] \leq 1$ holds for all t, Inequality (3.5) ensures the convergence of $E[w_t(B)]$ to

its supremum. It remains for us to show that this limit is equal to 1. Firstly, note that the random variable ρ_t can be rewritten in the following manner:

$$\rho_t = 1 + w_t(b)^{K-1}\left(\frac{1}{w_t(B) + w_t(b)\exp[\eta(b-B)\frac{Q}{K}]} - 1\right)1 + \frac{w_t(b)^K(1-\exp[\eta(b-B)\frac{Q}{K}])}{1 - w_t(b)(1-\exp[\eta(b-B)\frac{Q}{K}])}$$

$$= 1 + \frac{w_t(b)(1-A)}{1-w_t(b)(1-A)} = 1 + \frac{(1-w_t(B))^K(1-A)}{A + w_t(B)(1-A)}$$

Thus, we have:

$$\mathbb{E}[w_{t+1}(B)|w_t(B)] - w_t(B) = w_t(B)\frac{(1-w_t(B))^K(1-A)}{A + w_t(B)(1-A)} \tag{3.6}$$

By applying the law of total expectation to (3.6) we obtain:

$$\mathbb{E}[w_{t+1}(B)] - \mathbb{E}[w_t(B)] = \mathbb{E}\left[w_t(B)\frac{(1-w_t(B))^K(1-A)}{A + w_t(B)(1-A)}\right]$$

Since the sequence $(E[w_t(B)])_t$ converges it follows that:

$$\lim_{t\to\infty}\mathbb{E}[w_{t+1}(B)] - \mathbb{E}[w_t(B)] = \lim_{t\to\infty}\mathbb{E}\left[w_t(B)\frac{(1-w_t(B))^K(1-A)}{A + w_t(B)(1-A)}\right] = 0$$

Next, because $w_t(B)\frac{(1-w_t(B))^K(1-A)}{A+w_t(B)(1-A)} \leq 1$ for all t, we can swap the limit and the expectation which, together with the fact that $w_t(B)$ converge almost surely to $w_\infty(B)$, allows us to write:

$$\lim_{t\to\infty}\mathbb{E}\left[w_t(B)\frac{(1-w_t(B))^K(1-A)}{A + w_t(B)(1-A)}\right] = \mathbb{E}\left[w_\infty(B)\frac{(1-w_\infty(B))^K(1-A)}{A + w_\infty(B)(1-A)}\right] = 0$$

Given that $w_t(B) \in [0,1]$ for all t, the above expectation is 0 if and only if the following condition is satisfied:

$$w_\infty(B)\frac{(1-w_\infty(B))^K(1-A)}{A + w_\infty(B)(1-A)} = 0$$

As established in the preceding analysis of ρ_t, this quantity is zero only when $w_\infty(B) \in \{0,1\}$. However, $w_\infty(B)$ cannot be 0 since $E[w_\infty(B)] > E[w_0(B)] = w_0(B) > 0$. Therefore, $w_t(B)$ converges to 1 almost surely. □

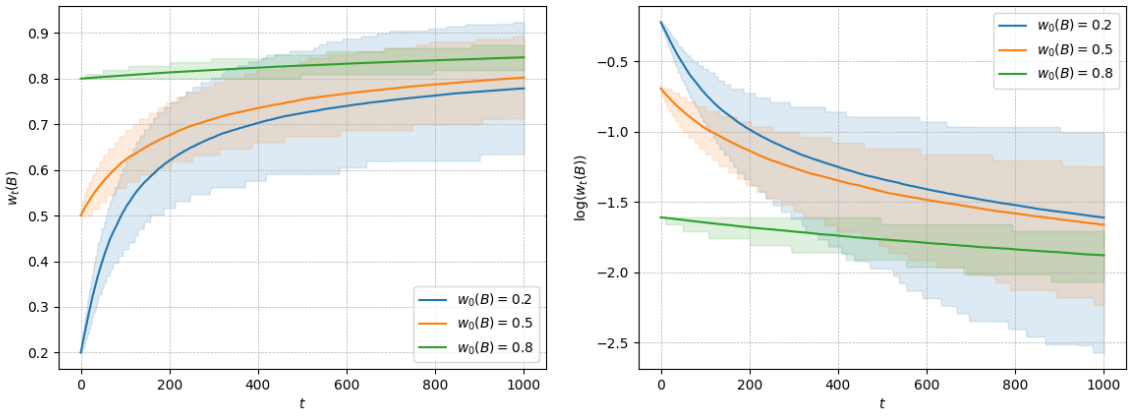


Figure 3.2: Left: plot of $w_t(B)$ against time for different starting values for an auction with the following characteristics: $Q = 11.484$, $T = 1000$, $(M, K) = (4, 4)$, $\overline{x} = \frac{Q}{K}$, $(B, b) = (1.1, 0.77)$ and $a = 0.7$. The solid line represents the mean, while the shaded areas cover the 90% confidence interval over 500 runs. Right: Same as the left plot, but we plot $\log w_t(B)$.

Figure 3.2 illustrates an example of the dynamics discussed in Theorem 4. As proved in the Theorem, $w_t(B)$ approaches 1 as t increases. Simultaneously, however, ρ_t also approaches 1. This simultaneous convergence explains why the curve grows in an exponential fashion for smaller values of $w_t(B)$, whereas it grows in a sublinear manner for larger values (as observed in the log plot on the right). Moreover, because of this simultaneous behavior, the confidence interval around the curves narrows as $w_0(B)$ increases.

The preceding result relied on the assumption that $\overline{x} = \frac{Q}{K}$. In what follows, we assume that the capacities lie in the interval $\frac{Q}{K} < \overline{x} < \frac{Q}{K-1}$ and prove that there exists a threshold $p \in (0, 1)$ such that for all $w_t(B) \in (p, 1)$ we are in a supermartingale regime.

Proposition 1. *If Inequality (3.2) does not hold,* $\frac{Q}{K} < \overline{x} < \frac{Q}{K-1}$*, and* $M = K$*, then for any non-trivial initial weight* $w_0(B) \in (0, 1)$ *there exists some* $p \in (0, 1)$ *such that:*

$$E[w_{t+1}(B)|w_t(B)] < w_t(B)$$

for all $w_t(B) \in (p, 1)$.

Proof. In order to show the existence of p and determine the function $E[w_{t+1}(B)|w_t(B)] - w_t(B)$ is negative in $(p, 1)$ it is sufficient to prove there exists an upper bound $U(w_t(B))$ of $E[w_{t+1}(B)|w_t(B)]$ such that $U(w_t(B)) - w_t$ has as root p and $U(w_t(B)) - w_t$ is negative on $(p, 1)$.

Since $(B-a)\frac{Q}{K} > (B-a)\frac{Q-(K-S-1)\overline{x}}{S+1}$ for all $S = 1, \dots, K-1$ we can upper bound the first term of Equation (3.3) in the following way:

$$\sum_{S=1}^{K-1} \binom{K-1}{S} w_t(b)^{K-S-1} w_t(B)^S \frac{w_t(B)\exp[\eta(B-a)\frac{Q-(K-S-1)\overline{x}}{S+1}]}{w_t(B)\exp[\eta(B-a)\frac{Q-(K-S-1)\overline{x}}{S+1}] + w_t(b)\exp[\eta(B-a)\overline{x}]}$$

$$< \frac{w_t(B)\exp[\eta(B-a)\frac{Q}{K}]}{w_t(B)\exp[\eta(B-a)\frac{Q}{K}] + w_t(b)\exp[\eta(B-a)\overline{x}]} \sum_{S=1}^{K-1} \binom{K-1}{S} w_t(b)^{K-S-1} w_t(B)^S$$

$$= \frac{w_t(B)\exp[\eta(B-a)\frac{Q}{K}]}{w_t(B)\exp[\eta(B-a)\frac{Q}{K}] + w_t(b)\exp[\eta(B-a)\overline{x}]} [1 - w_t(b)^{K-1}]$$

$$< \frac{w_t(B)\exp[\eta(B-a)\frac{Q}{K}]}{w_t(B)\exp[\eta(B-a)\frac{Q}{K}] + w_t(b)\exp[\eta(B-a)\overline{x}]}$$

Similarly, because $(B-a)\frac{Q}{K} > (B-a)(Q-(K-1)\overline{x})$, we can bound the second term of Equation (3.3) by:

$$w_t(b)^{K-1} \frac{w_t(B)\exp[\eta(B-a)(Q-(K-1)\overline{x})]}{w_t(B)\exp[\eta(B-a)(Q-(K-1)\overline{x})] + w_t(b)\exp[\eta(b-a)\frac{Q}{K}]}$$

$$< w_t(b)^{K-1} \frac{w_t(B)\exp[\eta(B-a)\frac{Q}{K}]}{w_t(B)\exp[\eta(B-a)\frac{Q}{K}] + w_t(b)\exp[\eta(b-a)\frac{Q}{K}]}$$

Therefore, by combining the two, we can bound the expectation by:

$$\begin{aligned} \mathbb{E}[w_{t+1}(B)|\mathcal{F}_t] < & \frac{w_t(B)\exp[\eta(B-a)\frac{Q}{K}]}{w_t(B)\exp[\eta(B-a)\frac{Q}{K}] + w_t(b)\exp[\eta(B-a)\overline{x}]} \\ & + w_t(b)^{K-1} \frac{w_t(B)\exp[\eta(B-a)\frac{Q}{K}]}{w_t(B)\exp[\eta(B-a)\frac{Q}{K}] + w_t(b)\exp[\eta(b-a)\frac{Q}{K}]} \end{aligned} \tag{3.7}$$

Finding a root of this upper bound is not tractable due to the fact that the term $w_t(b)$ is raised to the power of $K-1$. However, since $K \geq 2$ and $w_t(b) \in (0,1)$ we can bound $K-1$ by 2 for $K > 2$ and by 1 for $K = 2$.

1. **CASE** $K = 2$. In this case, the bound on the expectation becomes:

$$\mathbb{E}[w_{t+1}(B)|w_t(B)] < \frac{w_t(B)\exp[\eta(B-a)\frac{Q}{2}]}{w_t(B)\exp[\eta(B-a)\frac{Q}{2}] + w_t(b)\exp[\eta(B-a)\overline{x}]}$$

$$+ w_t(b)\frac{w_t(B)\exp[\eta(B-a)\frac{Q}{2}]}{w_t(B)\exp[\eta(B-a)\frac{Q}{2}] + w_t(b)\exp[\eta(b-a)\frac{Q}{2}]}$$

   By defining $A = \exp[\eta(B-a)\frac{Q}{2}]$, $B = \exp[\eta(B-a)\overline{x}]$ and $F = \exp[\eta(b-a)\frac{Q}{2}]$, we can rewrite the above bound as:

$$\mathbb{E}[w_{t+1}(B)|\mathcal{F}_t] < \frac{w_t(B)A}{w_t(B)A + w_t(b)D} + w_t(b)\frac{w_t(B)A}{w_t(B)A + w_t(b)F}$$

   The roots of this upper bound with respect to $w_t(B)$ satisfy the following equation:

$$\frac{w_t(B)A}{w_t(B)A + w_t(b)D} + w_t(b)\frac{w_t(B)A}{w_t(B)A + w_t(b)F} - w_t(B) = 0$$

   Via some standard simplifications, the left-hand side can be written as:

$$f(w_t(B)) = \frac{w_t(B)[1 - w_t(B)]g(w_t(B))}{[w_t(B)A + w_t(b)D] \cdot [w_t(B)A + w_t(b)F]}$$

   where:

$$g(w_t(B)) = w_t(B)(-2A^2 + 2AD + AF - DF) + DF - AD - AF$$

   Clearly, $f(w_t(B))$ has three roots: one at 0, one at 1, and the remaining one is the root of $g(w_t(B))$. Since $D > A > F$ we have $g(0) < 0$ and $g(1) > 0$. Therefore there exists a $p \in (0,1)$ such that $g(p) = 0$. This implies that:

$$\mathbb{E}[w_{t+1}(B)|w_t(B)] < \frac{w_t(B)A}{w_t(B)A + w_t(b)D} + w_t(b)\frac{w_t(B)A}{w_t(B)A + w_t(b)F} < w_t(B)$$

   for all $w_t(B) \in (p,1)$.

2. **CASE** $K > 2$. In this case can bound $w_t(b)^{k-1}$ by $w_t(b)^2$. Then the expectation can

be bounded by:

$$\mathbb{E}[w_{t+1}(B)|\mathcal{F}_t] < \frac{w_t(B)\exp[\eta(B-a)\frac{Q}{K}]}{w_t(B)\exp[\eta(B-a)\frac{Q}{K}] + w_t(b)\exp[\eta(B-a)\overline{x}]}$$

$$+ w_t(b)^2 \frac{w_t(B)\exp[\eta(B-a)\frac{Q}{K}]}{w_t(B)\exp[\eta(B-a)\frac{Q}{K}] + w_t(b)\exp[\eta(b-a)\frac{Q}{K}]}$$

As before we define $A = \exp[\eta(B-a)\frac{Q}{K}]$, $B = \exp[\eta(B-a)\overline{x}]$ and $F = \exp[\eta(b-a)\frac{Q}{K}]$. The roots of this upper bound with respect to $w_t(B)$ satisfy the following equation:

$$\frac{w_t(B)A}{w_t(B)A + w_t(b)D} + w_t(b)^2 \frac{w_t(B)A}{w_t(B)A + w_t(b)F} - w_t(B) = 0$$

The left-hand side can be written as:

$$f(w_t(B)) = \frac{w_t(B)[1-w_t(B)]g(w_t(B))}{[w_t(B)A + w_t(b)D] \cdot [w_t(B)A + w_t(b)F]}$$

where:

$$g(w_t(B)) = w_t(B)^2 A(A-D) + w_t(B)(-2A^2 + AF + 3AD - DF) + DF - AD - AF$$

Since $D > A > F$, the determinant is positive, and the parabola is downward opening. Therefore, we have 4 roots in total: the two trivial ones (i.e., 0 and 1) and, using the same reasoning as before, one root, p, in $(0,1)$. Because of the shape of the parabola and since $g(1) > 0$, the other root is outside the interval $[0,1]$. This implies that:

$$\mathbb{E}[w_{t+1}(B)|w_t(B)] < \frac{w_t(B)A}{w_t(B)A + w_t(b)D} + w_t(b)^2 \frac{w_t(B)A}{w_t(B)A + w_t(b)F} < w_t(B)$$

for all $w_t(B) \in (p,1)$.

Therefore in both cases there exists a $p \in (0,1)$ such that:

$$E[w_{t+1}(B)|\mathcal{F}_t] < w_t(B)$$

for all $w_t(B) \in (p,1)$. □

In the next result, we use Proposition 1 and the fact that $E[w_{t+1}(B)|w_t(B)] - w_t(B)$ is continuous in order to show that there exists some $p', p \in [0,1)$ with $p' < p$ such that for all

$w_t(B) \in (p, 1)$ we are in a supermartingale regime, p is a fixed point of $E[w_{t+1}(B)|w_t(B)] - w_t(B)$ and that in (p', p) we are in a submartingale regime [5].

Theorem 5. *If Inequality (3.2) does not hold,* $\frac{Q}{K} < \overline{x} < \frac{Q}{K-1}$, *and* $M = K$ *then for any non-trivial initial weight* $w_0(B) \in (0, 1)$ *there exists some* $p', p \in [0, 1)$ *with* $p' < p$ *such that:*

1. $E[w_{t+1}(B)|\mathcal{F}_t] < w_t(B)$ *for all* $w_t(B) \in (p, 1)$.
2. $E[w_{t+1}(B)|w_t(B)] = w_t(B)$ *when* $w_t(B) = p$.
3. $E[w_{t+1}(B)|\mathcal{F}_t] > w_t(B)$ *for all* $w_t(B) \in (p', p)$.

Proof. The first statement has been proven in Proposition 1. For the remaining two statements, define the function $f(w_t(B)) = \mathbb{E}[w_{t+1}(B)|\mathcal{F}_t] - w_t(B)$. Since f is continuous on $[0, 1]$ and it satisfies $f(0) = f(1) = 0$ while being strictly negative on $(p, 1)$, it must be zero at p. However, this does not rule out the existence of additional roots in the interior. Therefore, by continuity there exists a sub-interval $[p', p) \subset [0, p)$ where $E[w_{t+1}(B)|\mathcal{F}_t] > w_t(B)$ for all $w_t(B) \in (p', p)$. □

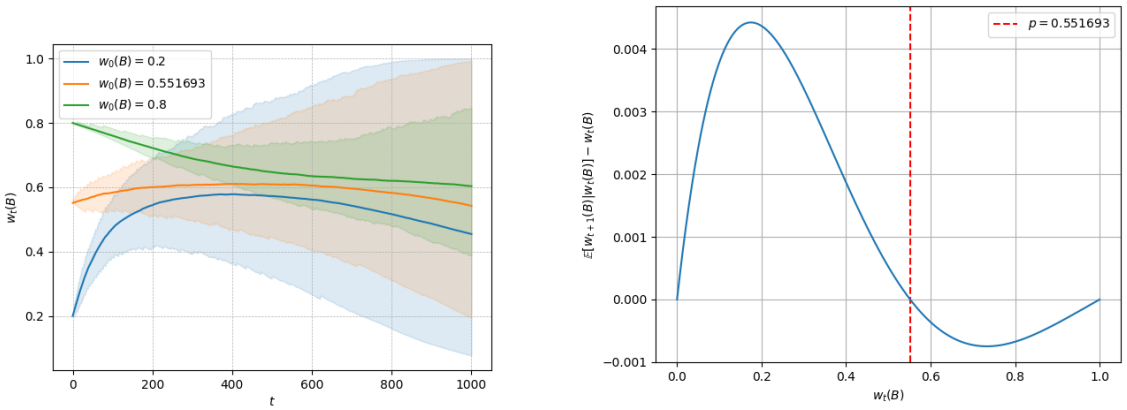


Figure 3.3: Left: Plot of $w_t(B)$ over time for different starting values for an auction with the following characteristics: $Q = 11.484$, $T = 1000$, $(M, K) = (4, 4)$, $\overline{x} = 3 > \frac{Q}{K}$, $(B, b) = (1.1, 0.77)$ and $a = 0.7$. The solid line represents the mean, while the shaded areas cover the 90% confidence interval over 500 runs. Right: The function $\mathbb{E}[w_{t+1}(B)|w_t(B)] - w_t(B)$ for the data just described.

Figure 3.3 illustrates an example of the dynamics discussed in Theorem 5. As shown in the right panel, we have a submartingale on $(0, p)$ and a supermartingale on $(p, 1)$.

[5] Unfortunately, we were not able to extend this result to unconditional expectation because we do not have any information on the behavior of the weights between 0 and p'.

This supermartingale behavior is clearly visible in the left-hand plot, where the green line (which starts above the root $p = 0.551693$) decreases as t increases. Conversely, the orange trajectory (which originates at the equilibrium point) does not remain stationary in our simulation. While this variance is to be expected since our results are based on (conditional) expectation, it raises questions concerning the stability of this equilibrium point. Lastly, the blue line initially approaches $y = p$ and subsequently decreases. This trend suggests that the individual paths that cross $y = p$ immediately encounter the downward pressure from the supermartingale.

This concludes the results using the assumption that $M = K$. A natural follow-up question is how the EW dynamics change when $M > K$. Unfortunately, deriving analytical bounds on the expectation proved intractable [6]. Nevertheless, we did conduct a numerical simulation that offers some insights.

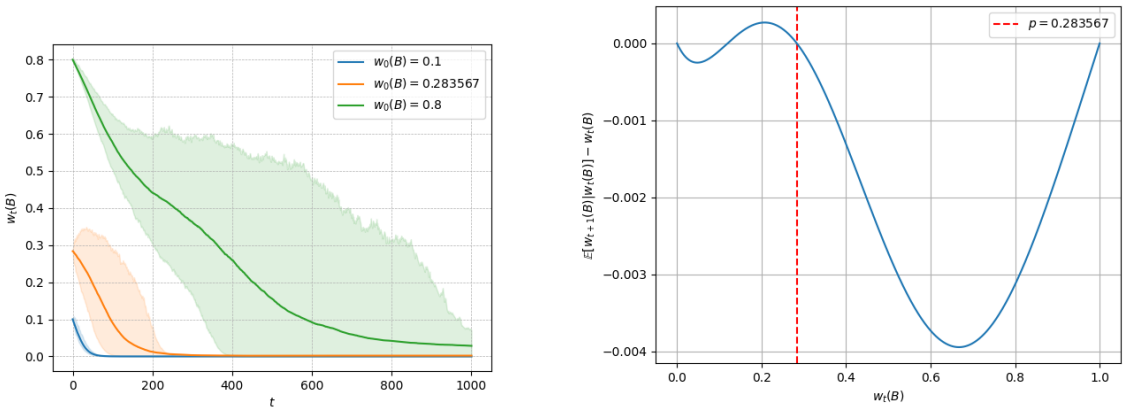


Figure 3.4: Left: plot of $w_t(B)$ over time for different starting values for an auction with the following characteristics: $Q = 11.484$, $T = 1000$, $(M, K) = (5, 4)$, $\overline{x} = \frac{Q}{K}$, $(B, b) = (1.1, 0.77)$ and $a = 0.7$. The solid line represents the mean, while the shaded areas cover the 90% confidence interval over 500 runs. Right: The function $\mathbb{E}[w_{t+1}(B)|w_t(B)] - w_t(B)$ for the data just described.

The scenario we consider in Figure 3.4 is the same as the one in Figure 3.2 except for the fact that $M = K + 1$. However, there are important differences. First of all, the function $\mathbb{E}[w_{t+1}(B)|w_t(B)] - w_t(B)$ did not have any roots in $(0, 1)$ in the previous case, but has 2 in this one. Next, in this figure, the weight $w_t(B)$ goes to 0 rather than approaching 1 as in Figure 3.2. This has an intuitive explanation, as, in this case, there is one additional

[6] This is because the conditional expectation is shaped like a logistic function with an inflection point and therefore results like convex or stochastic ordering could not be used.

scenario where players who bid B are paid out 0. This means that there is one additional case where a player would rather bid b rather than B, even though that means receiving a smaller payout. While we were not able to prove analytical results for $M > K$, we conjecture it is likely possible to show that we are in a supermartingale regime for a sufficiently large portion of $[0, 1]$ and therefore the expected value of $w_t(B)$ approaches 0.

One of the core assumptions of this section was that all players share the same capacities. This is not necessarily true because there may exist different players who have, for example, more generators than other players. However, if these capacities were distinct (but still in the range $\left[\frac{Q}{K}, \frac{Q}{K-1}\right)$) then the difference $q(S) - q(S+1)$ would not necessarily be positive and therefore we could not reduce the system of Inequalities (3.1) in the same way we have done before. To see this, consider the following:

$$\min_{\frac{Q}{K-1} > \overline{x} \geq \frac{Q}{K}} (q(S) - q(S+1)) = \frac{Q - S\frac{Q}{K-1}}{M - S} - \frac{Q - (S+1)\frac{Q}{K}}{M - (S+1)}$$

The numerator simplifies to:

$$Q[K(M - S - 1)(K - 1 - S) - (K - S - 1)(M - S)(K - 1)]$$

$$= Q(K - S - 1)(KM - KS - K - MK + M + SK - S) =$$

$$= Q(K - S - 1)(M - K - S)$$

Therefore:

$$\min_{\frac{Q}{K-1} > \overline{x} \geq \frac{Q}{K}} (q(S) - q(S+1)) = \frac{Q(K - S - 1)(M - K - S)}{K(K - 1)(M - S)(M - S - 1)}$$

Since $M \geq K \geq 2$ and $S \in \{0, \ldots, K - 2\}$ all terms of this fraction are always positive except $M - K - S$. To preserve the same inequality (from the point of view of player i) as in (3.2), we need $M - K - S$ to be positive for all S and, in particular, for the largest S, i.e., $K - 2$. Therefore, the additional constraint we have to consider in this case is $M \geq 2K - 2$. This means that if the number of players is not at least $2K - 2$, then it is possible that sharing energy with an additional player can be more beneficial (in terms of allocated energy) than sharing with just S players.

In the next chapter, we move away from the assumption of homogeneous players and address the case with two players who have different capacities and distinct linear costs.

4 Auction with two players, N bids and linear costs

In the previous chapter, we assumed that all players have identical characteristics, both from a production and a bidding standpoint. However, different producers possess unique capacities, cost functions (e.g., wind vs nuclear plants), and varying degrees of risk aversion, since they operate under diverse market and weather conditions. Moreover, they might hold divergent views regarding what the "true" price of energy at a given time should be, as well as different levels of certainty in this estimation. This ultimately influences both the quantity and the range of possible bids. Therefore, in this chapter, we study an auction with two players that satisfies the following assumptions:

1. Each player $i \in \{1, 2\}$ has a finite strategy set of the form $\mathcal{B}_i = \{b_1^{(i)}, \dots, b_N^{(i)}\}$ where the elements are ordered such that $b_j^{(i)} > b_k^{(i)}$ if $j > k$. We denote by $B_i = b_N^{(i)}$ and $b_i = b_1^{(i)}$ the largest and smallest bids respectively.
2. The capacities are bounded $\frac{Q}{2} < \overline{x}_1, \overline{x}_2 < Q$.
3. We consider two distinct costs functions denoted by $C_1(x)$ and $C_2(x)$ which satisfy the following properties:
   - $C_1(x) = ax$ with $b_1 > a$ so that even the smallest bid gives a profit.
   - $C_2(x) = dx$ with $b_2 > d$.

In the sections that follow, we study the dynamics of the EW algorithm under different scenarios, depending on the relationship between the largest and smallest bids in the two strategy sets. We are primarily interested in scenarios where the players do not share the same B_i or b_i. However, we still discuss the scenario in which $B_1 > b_1 = B_2 > b_2$, as it yields interesting theoretical results that can be applied to other cases. Therefore, excluding symmetric cases, we have 4 scenarios to analyze:

1. $B_1 > b_1 > B_2 > b_2$.

2. $B_1 > b_1 = B_2 > b_2$.

3. $B_1 > B_2 > b_1 > b_2$.

4. $B_1 > B_2 > b_2 > b_1$.

We proceed in this order.

4.1 Scenario 1: $B_1 > b_1 > B_2 > b_2$

Under this bidding order, there is only one possible limit. The following result shows that $w_t(B_1)$, independently of the starting weight (as long as it is not zero), approaches 1 as t approaches infinity. Moreover, while at the beginning of this chapter we have stated that cost functions are linear, the following statement is true for any cost function. Therefore, in the proof, we simply refer to the cost function as $C_1(x)$.

Theorem 6. *If we are in Scenario 1, then, for any non-trivial initial initialization $w_0(B_1) \in (0,1)$, we have $w_t(B_1) \xrightarrow[t\to\infty]{} 1$.*

Proof. First of all note that for two arbitrary bids $b_j^{(1)} \in \mathcal{B}_1$ and $b_k^{(2)} \in \mathcal{B}_2$ we have $b_j^{(1)} > b_k^{(2)}$. Therefore, if we solve the market-clearing problem, the utilities of the two players are:

$$u_1(b_j^{(1)}, b_k^{(2)}) = b_j^{(1)}(Q - \overline{x}_2) - C_1(Q - \overline{x}_2) \quad \text{and} \quad u_2(b_j^{(1)}, b_k^{(2)}) = b_j^{(1)}\overline{x}_2 - C_2(\overline{x}_2)$$

This implies that player 1 always sets the market-clearing price, and its payoff function is always of the same form. Because of this at $t = 1$ the weight of bid B_1 is:

$$w_1(B_1) = \frac{w_0(B_1) \cdot e^{\eta(B_1 x_1^* - C_1(x_1^*))}}{\sum_{j=1}^N w_0(b_j^{(1)}) \cdot e^{\eta(b_j^{(1)} x_1^* - C_1(x_1^*))}} = \frac{w_0(B_1) e^{\eta B_1 x_1^*}}{\sum_{j=1}^N w_0(b_j^{(1)}) e^{\eta b_j^{(1)} x_1^*}}$$

At $t = 2$ we have:

$$w_2(B_1) = \frac{w_1(B_1) \cdot e^{\eta(B_1 x_1^* - C_1(x_1^*))}}{\sum_{j=1}^N w_1(b_j^{(1)}) \cdot e^{\eta(b_j^{(1)} x_1^* - C_1(x_1^*))}} = \frac{w_1(B_1) e^{\eta B_1 x_1^*}}{\sum_{j=1}^N w_1(b_j^{(1)}) e^{\eta b_j^{(1)} x_1^*}} =$$

$$= \frac{\frac{w_0(B_1) e^{\eta B_1 x_1^*}}{\sum_{k=1}^N w_0(b_k^{(1)}) e^{\eta b_k^{(1)} x_1^*}} e^{\eta B_1 x_1^*}}{\sum_{j=1}^N \frac{w_0(b_j^{(1)}) e^{\eta b_j^{(1)} x_1^*}}{\sum_{k=1}^N w_0(d_k^{(1)}) e^{\eta d_k^{(1)} x_1^*}} e^{\eta b_j^{(1)} x_1^*}} = \frac{w_0(B_1) e^{2\eta B_1 x_1^*}}{\sum_{j=1}^N w_0(b_j^{(1)}) e^{2\eta b_j^{(1)} x_1^*}}$$

Continuing this reasoning at $t = T$ we have:

$$w_T(B_1) = \frac{w_0(B_1)e^{T\eta B_1 x_1^*}}{\sum_{j=1}^{N} w_0(b_j^{(1)})e^{T\eta b_j^{(1)} x_1^*}}$$

Now consider $1 - w_T(B_1)$. We can write (keeping in mind that $d_N^{(1)} = B_1$ according to our convention):

$$1 - w_T(B_1) = 1 - \frac{w_0(B_1)e^{T\eta B_1 x_1^*}}{\sum_{j=1}^{N} w_0(b_j^{(1)})e^{T\eta b_j^{(1)} x_1^*}} = \frac{\sum_{j=1}^{N-1} w_0(b_j^{(1)})e^{T\eta b_j^{(1)} x_1^*}}{\sum_{j=1}^{N} w_0(b_j^{(1)})e^{T\eta b_j^{(1)} x_1^*}}$$

Note that we can bound the numerator by:

$$\sum_{j=1}^{N-1} w_0(b_j^{(1)})e^{T\eta b_j^{(1)} x_1^*} < \sum_{j=1}^{N-1} e^{T\eta b_j^{(1)} x_1^*} < \sum_{j=1}^{N-1} e^{T\eta b_{(N-1)}^{(1)} x_1^*} = (N-1)e^{T\eta b_{(N-1)}^{(1)} x_1^*}$$

We can bound the denominator by;

$$\sum_{j=1}^{N} w_0(b_j^{(1)})e^{T\eta b_j^{(1)} x_1^*} \geq w_0(b_N^{(1)})e^{T\eta b_N^{(1)} x_1^*}$$

Combining the two bounds, we obtain:

$$1 - w_T(B_1) < \frac{(N-1)e^{T\eta b_{(N-1)}^{(1)} x_1^*}}{w_0(b_N^{(1)})e^{T\eta b_N^{(1)} x_1^*}} = \frac{N-1}{w_0(b_N^{(1)})} e^{T\eta(b_{(N-1)}^{(1)} - b_{(N)}^{(1)})x_1^*}$$

Since $b_{(N)}^{(1)} > b_{(N-1)}^{(1)}$ it follows that;

$$\lim_{T\to\infty} 1 - w_T(B_1) = \lim_{T\to\infty} \frac{N-1}{w_0(b_N^{(1)})} e^{T\eta(b_{(N-1)}^{(1)} - b_{(N)}^{(1)})x_1^*} = 0$$

from which we can conclude that the weight associated with B_1 converges to 1. □

Thus, the sequence $w_T(B_1)$ converges to 1 at a rate of $\mathcal{O}(e^{T\eta(b_{(N-1)}^{(1)} - b_{(N)}^{(1)})x_1^*})$. This implies that, if the distance between B_1 and the second largest bid increases, the sequence converges faster.

Remark. *From the observation made at the beginning of the proof of the Theorem 6, we can clearly see that player 2 never sets the clearing price, and therefore, submitting another*

bid does not change the payoff of player 2. Therefore, the weights of player 2 in this scenario do not change over time.

4.2 Scenario 2: $B_1 > b_1 = B_2 > b_2$

While this scenario is very similar to the first one, depending on the utility gained when the two players tie (i.e., when player 1 bids b_1 and player 2 bids B_2), there can be two different convergence types. Table 4.1 displays the payoff matrix of the auction seen as a bimatrix game where $f(z) = (z-a)(Q-\overline{x}_2)$, $g(z) = (z-d)\overline{x}_2$ and $K = b_1 = B_2$.

	$b_1^{(2)}$	$\dots$	$b_{N-1}^{(2)}$	$b_N^{(2)} := K$
$b_N^{(1)}$	$u_1 = f(b_N^{(1)})$ $u_2 = g(b_N^{(1)})$	$\dots$	$u_1 = f(b_N^{(1)})$ $u_2 = g(b_N^{(1)})$	$u_1 = f(b_N^{(1)})$ $u_2 = g(b_N^{(1)})$
$b_{N-1}^{(1)}$	$u_1 = f(b_{N-1}^{(1)})$ $u_2 = g(b_{N-1}^{(1)})$	$\dots$	$u_1 = f(b_{N-1}^{(1)})$ $u_2 = g(b_{N-1}^{(1)})$	$u_1 = f(b_{N-1}^{(1)})$ $u_2 = g(b_{N-1}^{(1)})$
$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$
$b_1^{(1)} := K$	$u_1 = f(b_1^{(1)})$ $u_2 = g(b_1^{(1)})$	$\dots$	$u_1 = f(b_1^{(1)})$ $u_2 = g(b_1^{(1)})$	$u_1 = K\frac{Q}{2} - C(\frac{Q}{2})$ $u_2 = K\frac{Q}{2} - C(\frac{Q}{2})$

Table 4.1: Payoff Matrix Scenario 2

For each player, the utilities have the following order:

- $f(b_N^{(1)}) > f(b_{N-1}^{(1)}) > \dots > f(b_1^{(1)})$ and $(K-a)\frac{Q}{2} > (K-a)(Q-\overline{x}_2) = f(b_1^{(1)})$.
- $g(b_N^{(1)}) > g(b_{N-1}^{(1)}) > \dots > g(b_1^{(1)})$ and $g(b_1^{(1)}) = (K-d)\overline{x}_2 > (K-d)\frac{Q}{2}$.

Remark (Order matrix)**.** *In what follows, the actual values of the payoffs are irrelevant: what matters is the order between them. Therefore, we are going to use an order matrix where each entry is a number between* 0 *and* N*; where* 0 *is the smallest payoff while* N *is the largest.*

Therefore, the utilities of player 2 can be uniquely ordered, while the utilities of player 1 can be ordered in $N+1$ different ways.

In the following result, we show that if we are in one of the N cases in which $(K-a)\frac{Q}{2} < f(b_N^{(1)})$ (where bidding B_1 always leads to a higher payoff) then $w_t(B)$ approaches 1 as t approaches infinity.

Theorem 7. *If we are in Scenario 2 and and* $(K-a)\frac{Q}{2} < f(b_N^{(1)})$*, then, for any non-trivial initial weight* $w_0(B_1)$*, we have* $w_t(B_1) \xrightarrow[t\to\infty]{} 1$*.*

Proof. Assume that $K\frac{Q}{2} - C(\frac{Q}{2}) < f(b_N^{(1)})$. The order matrix is:

	$b_1^{(2)}$	...	$b_{N-1}^{(2)}$	$b_N^{(2)} := K$
$b_N^{(1)}$	(N,N)	...	(N,N)	(N,N)
$b_{N-1}^{(1)}$	(r,N-1)	...	(r,N-1)	(r,N-1)
$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$b_1^{(1)} := K$	(s,1)	...	(s,1)	(t,0)

where we put r, s, t (with $s < t < N$) as some ranks of the payoffs as placeholders (their relations do not matter). By looking at this order matrix, we can clearly see that player 2 has only an incentive to switch bids when both players bid K. In all other cases, there is no incentive as the payoffs are constant. However, player 1 will always prefer B_1 as this always leads to the highest payoff no matter what player 2 bids. This implies that, like in scenario 1 (albeit with a slightly different convergence rate), the weight associated with B_1 will converge to 1. □

Remark. *Unlike Scenario 1, where player 2 never sets the market-clearing price, when both players bid* K*, then player 2 does set the price (together with player 1). Therefore, the weights of player 2 will remain constant as long as player 1 does not bid* K*. In particular, since* $w_t(B_1)$ *approaches 1 it follows that* $w_t(K)$ *goes to 0. Once that happens, the weights of player 2 will stall. However, we do not know this time a priori.*

In the last case, i.e where $(K-a)\frac{Q}{2} > f(b_N^{(1)}) = f(B_1)$, we cannot conclude the same deterministic convergence as before. Instead, the following result demonstrates that $\mathbb{E}[w_t(B)]$ approaches 1 as t approaches infinity. Because the proof is quite involved, we first show it for $N = 2$ and then, in Theorem 8, we show it for an arbitrary N.

Proposition 2. *If we are in Scenario 2 with* $N = 2$ *and* $(K-a)\frac{Q}{2} > f(b_N^{(1)})$ *then, for any non-trivial initial weights* $w_0(B_1), w_0(B_2) \in (0,1)$*, we have* $\mathbb{E}[w_t(B_1)] \xrightarrow[t\to\infty]{} 1$*.*

Proof. In this case, the order matrix is:

	b_2	$B_2 = K$
B_1	(1,2)	(1,2)
$b_1 = K$	(0,1)	(2,0)

Let $①_{Pi}, ②_{Pi}, ③_{Pi}, ④_{Pi}$ be the utilities of player i in quadrant 1,2,3 and 4 respectively in the usual cartesian order. Define two variables:

$$Z_1^t = \ln\left[\frac{w_t(B_1)}{w_t(b_1)}\right] \qquad \text{and} \qquad Z_2^t = \ln\left[\frac{w_t(B_2)}{w_t(b_2)}\right]$$

Note that we have the following equality:

$$Z_1^{t+1} = \ln\left[\frac{w_{t+1}(B_1)}{w_{t+1}(b_1)}\right] = \ln\left[\frac{w_t(B_1)e^{\eta u(B_1,y_2^t)}}{w_t(b_1)e^{\eta u(b_1,y_2^t)}}\right]$$

where y_2^t is the action played by player 2 at time t. Then we have:

$$Z_1^{t+1} - Z_1^t = \eta\left[u(B_1, y_2^t) - u(b_1, y_2^t)\right]$$

Similarly:

$$Z_2^{t+1} - Z_2^t = \eta\left[u(y_1^t, B_2) - u(y_1^t, b_2)\right]$$

where y_1^t is the action played by player 1 at time t. Note that:

$$Z_2^{t+1} - Z_2^t\Big|_{y_1^t=B_1} = \eta[②_{P2} - ①_{P2}] = 0$$

and:

$$Z_2^{t+1} - Z_2^t\Big|_{y_1^t=b_1} = \eta[④_{P2} - ③_{P2}] < 0$$

From this, we can easily see that :

$$Z_2^{t+1} \leq Z_2^t \qquad \text{or equivalently} \qquad w_t(b_2) \leq w_{t+1}(b_2)$$

It follows that $w_t(b_2)$ is a submartingale. Moreover, since $w_t(b_2) \in [0,1]$ for all t, Theorem 3 guarantees that $w_t(b_2)$ converges almost surely to a variable $w_\infty(b_2)$ with finite mean. Let us consider now the random variable $Z_1^{t+1} - Z_1^t$ for the two possible actions of player 2.

$$Z_1^{t+1} - Z_1^t\Big|_{y_2^t=b_2} = \eta[①_{P1} - ③_{P1}] = \eta[(B_1 - a)(Q - \overline{x}_2) - (b_1 - a)(Q - \overline{x}_2)] = \eta(B_1 - b_1)(Q - \overline{x}_2)$$

and

$$Z_1^{t+1} - Z_1^t\Big|_{y_2^t=B_2} = \eta[②_{P1} - ④_{P1}] = \eta[(B_1 - a)(Q - \overline{x}_2) - (K - a)\frac{Q}{2}]$$

Therefore the expectation with respect to filtration $\mathcal{F}_t$ is:

$$\mathbb{E}[Z_1^{t+1} - Z_1^t | \mathcal{F}_t] = w_t(b_2)\eta(B_1 - K)(Q - \overline{x}_2) + w_t(B_2)\eta[(B_1 - a)(Q - \overline{x}_2) - (K - a)\frac{Q}{2}]$$

$$= w_t(b_2)\eta(K - a)(\overline{x}_2 - \frac{Q}{2}) + \eta[(B_1 - a)(Q - \overline{x}_2) - (K - a)\frac{Q}{2}]$$

If we now apply the law of total expectation, we obtain:

$$\mathbb{E}[Z_1^{t+1} - Z_1^t] = \mathbb{E}[w_t(b_2)]\eta(K - a)(\overline{x}_2 - \frac{Q}{2}) + \eta[(B_1 - a)(Q - \overline{x}_2) - (K - a)\frac{Q}{2}] \tag{4.1}$$

Note that since $w_t(b_2) \in [0, 1]$ it follows that $\lim_{t\to\infty} \mathbb{E}[w_t(b_2)] = \mathbb{E}[w_\infty(b_2)]$. Therefore, if we take the limit with respect to t in Equation (4.1) we obtain:

$$\lim_{t\to\infty} \mathbb{E}[Z_1^{t+1} - Z_1^t] = \mathbb{E}[w_\infty(b_2)]\eta(K - a)(\overline{x}_2 - \frac{Q}{2}) + \eta[(B_1 - a)(Q - \overline{x}_2) - (K - a)\frac{Q}{2}]$$

Since the term on the right-hand side is a fixed constant which is larger than 0 it follows that $\lim_{t\to\infty} \mathbb{E}[Z_1^t] \to \infty$. This happens if and only if $\mathbb{E}[w_t(b_1)] \to 0$ or equivalently if $\mathbb{E}[w_t(B_1)] \to 1$. □

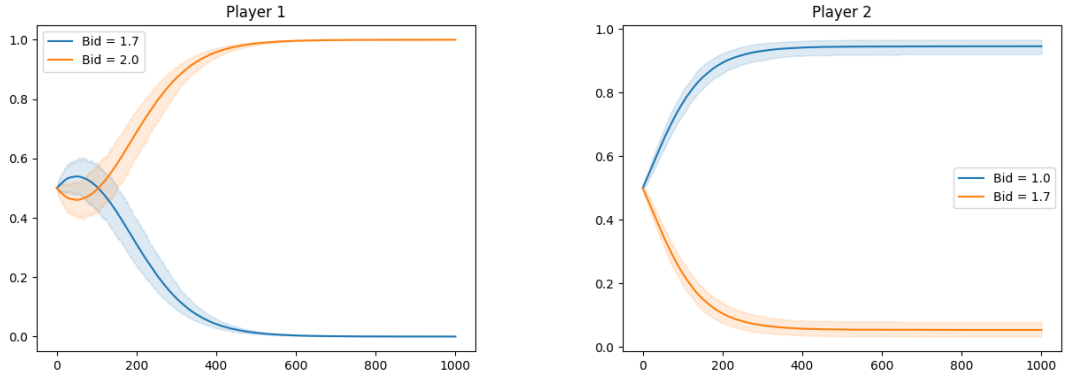


Figure 4.1: Evolution of the weights of the bids of player 1 (left) and player 2 (right) generated by running 1000 auctions with the following characteristics: $Q = 11.484$, $(B_1, b_1) = (2, 1.7)$, $a = 0.8$, $\overline{x}_1 = 7$, $(B_2, b_2) = (1.7, 1)$, $\overline{x}_2 = 9$ and $d = 0.7$. The solid line represents the mean, while the shaded areas cover the 90% confidence interval over 500 runs.

Figure 4.1 illustrates the case discussed in Proposition 2. We may notice that, as long as

player 2 bids K often enough, player 1 is incentivised to bid K as well because that gives the highest satisfaction (payoff). Therefore, initially, $w_t(b_1)$ increases. Simultaneously, $w_t(b_2)$ also increases because player 2 would rather receive a higher payoff. This proceeds until $w_t(b_2)$ reaches a threshold, above which player 1 is better off, in expectation, to bid B_1.

As mentioned at the beginning of this section, this result also extends to the case for arbitrary N by slightly tweaking the random variables Z_1^t, Z_1^t defined in Proposition 2.

Theorem 8. *If we are in Scenario 2 and* $(K-a)\frac{Q}{2} > f(b_N^{(1)})$ *then, for any non-trivial initial weights* $w_0(B_1), w_0(B_2) \in (0,1)$*, we have* $\mathbb{E}[w_t(B_1)] \xrightarrow[t\to\infty]{} 1$.

Proof. In this case, the payoff matrix is:

	$b_1^{(2)}$	...	$b_{N-1}^{(2)}$	$b_N^{(2)} := K$
$b_N^{(1)}$	(N-1,N)	...	(N-1,N)	(N-1,N)
$b_{N-1}^{(1)}$	(N-2,N-1)	...	(N-2,N-1)	(N-2,N-1)
$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$b_1^{(1)} := K$	(0,1)	...	(0,1)	(N,0)

Table 4.2: Case 2 bimatrix game

Similarly to the previous proof, we define the following random variables:

$$Z_2^t = \ln\left[\frac{w_t(b_N^{(2)})}{\sum_{i=1}^{N-1} w_t(b_i^{(2)})}\right] = \ln\left[\frac{w_{t-1}(b_N^{(2)})e^{\eta u(y_1^{t-1}, b_N^{(2)})}}{\sum_{i=1}^{N-1} w_{t-1}(b_i^{(2)})e^{\eta u(y_1^{t-1}, b_i^{(2)})}}\right]$$

where y_1^t is the action played by player 1 at time t. Then we have:

$$Z_2^{t+1} - Z_2^t = \ln\left[\frac{e^{\eta u(y_1^t, b_N^{(2)})}\sum_{i=1}^{N-1} w_t(b_i^{(2)})}{\sum_{i=1}^{N-1} w_t(b_i^{(2)})e^{\eta u(y_1^t, b_i^{(2)})}}\right]$$

For all y_1^t except $b_1^{(1)}$ we have equal payoffs and therefore:

$$Z_2^{t+1} - Z_2^t\Big|_{y_1^t \in \{b_2^{(1)}, \ldots, b_N^{(1)}\}} = \ln 1 = 0$$

On the other hand for $y_1^t = b_1^{(1)} = K$ we have:

$$Z_2^{t+1} - Z_2^t\Big|_{y_1^t = b_1^{(1)}} = \ln\left[\frac{e^{\eta(K\frac{Q}{2} - C(\frac{Q}{2}))}\sum_{i=1}^{N-1} w_t(b_i^{(2)})}{\sum_{i=1}^{N-1} w_t(b_i^{(2)})e^{\eta(K\overline{x}_2 - C(\overline{x}_2)}}\right] =$$

$$= \eta(K-a)(\frac{Q}{2} - \overline{x}_2) < 0$$

Therefore, as before we have, $Z_2^{t+1} \leq Z_2^t$ or equivalently $\sum_{i=1}^{N-1} w_t(b_i^{(2)}) \leq \sum_{i=1}^{N-1} w_{t+1}(b_i^{(2)})$. From which it follows that $\sum_{i=1}^{N-1} w_{t+1}(b_i^{(2)})$ is a submartingale. Moreover, since $\sum_{i=1}^{N-1} w_{t+1}(b_i^{(2)}) \in [0,1]$ for all t, Theorem 3 guarantees that $\sum_{i=1}^{N-1} w_{t+1}(b_i^{(2)})$ converges almost surely to a random variable with finite mean. Next, similarly to what was done in the $N = 2$ case, we define:

$$Z_1^t = \ln\left[\frac{\sum_{i=2}^{N} w_t(b_i^{(1)})}{w(b_1^{(1)})}\right]$$

and we have:

$$Z_1^{t+1} - Z_1^t = \ln\left[\frac{\sum_{i=2}^{N} w_t(b_i^{(1)}) e^{\eta u(b_i^{(1)}, y_2^t)}}{e^{\eta u(b_1^{(1)}, y_2^t)} \sum_{i=2}^{N} w_t(b_i^{(1)})}\right]$$

where y_2^t is the action played by player 2 at time t. The expectation of this difference can be written (similarly as in the $N = 2$ case) in the following way:

$$\mathbb{E}[Z_1^{t+1} - Z_1^t | \mathcal{F}_t] = \alpha \sum_{i=1}^{N-1} w_t(b_i^{(2)}) + \beta$$

for $\alpha > 0$. Therefore, by repeating the same reasoning that was done for the case $N = 2$, we have $\mathbb{E}[w_t(B)] \xrightarrow[t\to\infty]{} 1$ □

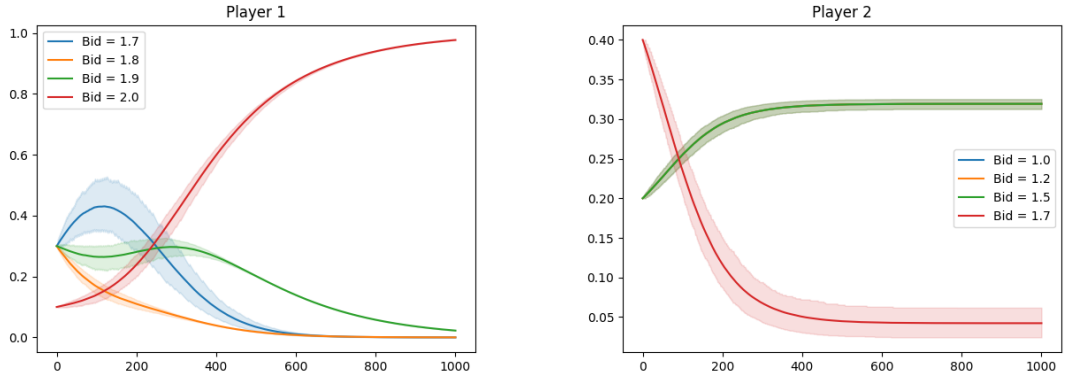


Figure 4.2: Evolution of the weights of the bids of player 1 (left) and player 2 (right) generated by running 1000 auctions with the following characteristics: $Q = 11.484$, $\mathcal{B}_1 = \{1.7, 1.8, 1.9, 2\}$, $\overline{x}_1 = 7$, $a = 0.8$, $\mathcal{B}_2 = \{1, 1.2, 1.5, 1.7\}$, $\overline{x}_2 = 9$ and $d = 0.7$. The solid line represents the mean, while the shaded areas cover the 90% confidence interval over 500 runs.

The behavior of the weights in Figure 4.2 is, of course, similar to that of Figure 4.1. However, there are some interesting insights that are worth discussing. First of all, on the left plot, we can see that $w(1.9)$ (i.e., the weight of the second largest bid) also starts increasing once $w_t(b_1)$ reaches its maximum. However, it decreases again as $w(B_1)$ keeps increasing since B_1 is the bid that guarantees the highest payoff. On the other hand, for player 2, all the weights (except the one associated with B_2) are overlapping (including the confidence intervals). This is to be expected, as if we fix any bid of player 1, the payoff gained by player 2 when they execute either of these bids is the same.

4.3 Scenario 3: $B_1 > B_2 > b_1 > b_2$

Although this section presents only a minor variation from Scenario 2, its analysis is considerably more complex due to the absence of (some) regularity in the payoffs' ordering. Consequently, the analytical results are restricted to $N = 2$. In particular, we obtain 4 subcases, and we show that we converge to either a NEEP or to a Strict Nash equilibrium. The general payoff matrix is:

	b_2	B_2
B_1	$u_1 = (B_1 - a)(Q - \overline{x}_2)$ $u_2 = (B_1 - d)\overline{x}_2$	$u_1 = (B_1 - a)(Q - \overline{x}_2)$ $u_2 = (B_1 - d)\overline{x}_2$
b_1	$u_1 = (b_1 - a)(Q - \overline{x}_2)$ $u_2 = (b_1 - d)\overline{x}_2$	$u_1 = (B_2 - a)\overline{x}_1$ $u_2 = (B_2 - d)(Q - \overline{x}_1)$

From these relations, the following orders can be deduced:

- $①_{P1} = ②_{P1} > ③_{P1}$ and $④_{P1} > ③_{P1}$.
- $①_{P2} = ②_{P2} > ③_{P2}$ and $①_{P2} = ②_{P2} > ④_{P2}$

These orders, in turn, yield the following cases:

1. $④_{P1} > ①_{P1} = ②_{P1} > ③_{P1}$ and $①_{P2} = ②_{P2} > ④_{P2} > ③_{P2}$
2. $④_{P1} > ①_{P1} = ②_{P1} > ③_{P1}$ and $①_{P2} = ②_{P2} > ③_{P2} > ④_{P2}$
3. $①_{P1} = ②_{P1} > ④_{P1} > ③_{P1}$ and $①_{P2} = ②_{P2} > ④_{P2} > ③_{P2}$
4. $①_{P1} = ②_{P1} > ④_{P1} > ③_{P1}$ and $①_{P2} = ②_{P2} > ③_{P2} > ④_{P2}$

	b_2	B_2
B_1	(1,2)	(1,2)
b_1	(0,0)	(2,1)

Table 4.3: Case 1

	b_2	B_2
B_1	(1,2)	(1,2)
b_1	(0,1)	(2,0)

Table 4.4: Case 2

	b_2	B_2
B_1	(2,2)	(2,2)
b_1	(0,0)	(1,1)

Table 4.5: Case 3

	b_2	B_2
B_1	(2,2)	(2,2)
b_1	(0,1)	(1,0)

Table 4.6: Case 4

Convergence for these cases can be easily deduced.

- CASE 1: By using the same proof style as in Proposition 2, we can show convergence to the Strict Nash Equilibrium ④.

- CASE 2: Since we have the same order matrix as in Proposition 2, the same convergence applies.
- CASE 3,4: As in Theorem 7.

Unfortunately, introducing additional bids noticeably increases the number of cases to consider. While we are not able to derive convergence guarantees, we present exploratory observations that provide a likely path toward understanding the weights' dynamics with N actions. In particular, we show that, provided the new bids are distinct from those of the other player [1], the bids that actively contribute to the dynamics are B_1 and B_2.

- Player 1. Consider an additional bid b_*.
  - If $B_1 > b_* > B_2$ then $b_* > y$ (where y is any pure action of player 2) and thus $u(b_*, y) = (b_* - a)(Q - \overline{x}_2)$. Therefore, in this case b_* behaves like a "weak" version of B_1.
  - If $B_2 > b_* > b_1$ then:

$$u_1(b_*, y) = (b_* - a)(Q - \overline{x}_2) - C(Q - \overline{x}_2) \quad \text{if} \quad b_* > y$$

$$u_1(b_*, y) = (y - d)\overline{x}_1 \quad \text{if} \quad b_* < y$$

    In the first case, b_* behaves like a weak B_1 while in the second case, Player 1's bid does not affect the setting of the MCP.

- Player 2. Consider an additional bid b_*.
  - If $B_2 > b_* > b_1$ then:

$$u_2(y, b_*) = (y - d)C(\overline{x}_2) \quad \text{if} \quad y > b_*$$

$$u_2(a_1, d) = (b_* - d)(Q - \overline{x}_1) \quad \text{if} \quad y < b_*$$

    In the first case, Player 2's bid does not affect the setting of the MCP, while in the second case, it behaves like a weak B_2.
  - If $b_1 > b_* > b_2$ then $u_2(y, b_*) = (y - d)\overline{x}_2$. Thus, in this case, it also has no effect.

[1] Identical bids would further increase the cases to consider due to the equal splitting of demand

4.4 Scenario 4: $B_1 > B_2 > b_2 > b_1$

Similarly to Scenario 3, we are not able to derive convergence guarantees for $N > 2$. Consequently, the analytical results are restricted to $N = 2$. In particular, we obtain 3 subcases, and we show that either converges to a NEEP or to a Strict Nash equilibrium. In this case, the payoff matrix is given by:

	b_2	B_2
B_1	$u_1 = (B_1 - a)(Q - \overline{x}_2)$ $u_2 = (B_1 - d)\overline{x}_2$	$u_1 = (B_1 - a)(Q - \overline{x}_2)$ $u_2 = (B_1 - d)\overline{x}_2$
b_1	$u_1 = (b_2 - a)\overline{x}_1$ $u_2 = (b_2 - d)(Q - \overline{x}_1)$	$u_1 = (B_2 - a)\overline{x}_1$ $u_2 = (B_2 - d)(Q - \overline{x}_1)$

The payoffs' relations are these:

- $①_{P2} = ②_{P2} > ④_{P2} > ③_{P2}$
- $④_{P1} > ③_{P1}$

From these relations, we deduce the following orders:

1. $①_{P1} = ②_{P1} > ④_{P1} > ③_{P1}$ and $①_{P2} = ②_{P2} > ④_{P2} > ③_{P2}$
2. $④_{P1} > ①_{P1} = ②_{P1} > ③_{P1}$ and $①_{P2} = ②_{P2} > ④_{P2} > ③_{P2}$
3. $④_{P1} > ③_{P1} > ①_{P1} = ②_{P1}$ and $①_{P2} = ②_{P2} > ④_{P2} > ③_{P2}$

	b_2	B_2
B_1	(2,2)	(2,2)
b_1	(0,0)	(1,1)

Table 4.7: Case 1

	b_2	B_2
B_1	(1,2)	(1,2)
b_1	(0, 0)	(2,1)

Table 4.8: Case 2

	b_2	B_2
B_1	(0,2)	(0,2)
b_1	(1,0)	(2,1)

Table 4.9: Case 3

Convergence results follow straightforwardly from the previous analysis.

- CASE 1: As in Theorem 7.
- CASES 2,3: By using the same proof style as in Proposition 2, we can show convergence to the Strict Nash Equilibrium ④.

We omit the detailed reasoning for the additional bid, as the underlying logic mirrors that of Scenario 2.

5 Conclusion

In this thesis, we established new convergence properties for the EW algorithm in both homogeneous and non-homogeneous settings. In the former, we derived a set of inequalities governing bidding preferences and showed that these collapse to a single inequality when cost functions are linear. Moreover, when this inequality is violated, we prove two theorems: the former describes the asymptotic dynamics when the aggregated capacities are equal to the demand, while the latter describes the local dynamics under a different capacity constraint. Next, we considered the more realistic case with non-homogeneous players with distinct costs and capacities who can choose from N distinct bids. In this setting, we identified four scenarios and proved convergence of the EW algorithm with $N \geq 2$ bids for the first two scenarios and proved convergence for $N = 2$ in the last two scenarios.

5.1 Future work

While several extensions are possible, we outline our immediate research priorities:

- In Theorem 5, we only showed what happens locally and conditional on a path. Is it possible to show what the asymptotic dynamics of $\mathbb{E}[w_t(B)]$ are? Can convergence to p be shown?
- At the end of Section 3.2, we provided a simulation that indicates that as M increases (while still violating Inequality (3.2)), $\mathbb{E}[w_t(B)]$ approaches zero. Can it be shown that this behavior is always true?
- Can it be shown what the asymptotic dynamics are in Scenarios 3 and 4 when the number of bids is above 2? Is it possible to determine convergence by simply considering the largest and smallest bids in light of the observation that we made?
- In this work, we focused on players that share similar capacities (at least belonging to the same interval). However, in reality, some energy producers may be able to generate quantities that are significantly larger than those of other players. What would the dynamics be under this "dominance"?

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